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JEE MAIN · CHEMISTRY · CHAPTER-WISE PYQS

# p-Block Elements

JEE Main Chemistry · Previous Year Questions with Answer Key

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## MCQ (Single Correct Answer) 121 questions

**Q1.** JEE Main 2026 (Online) 8th April Evening Shift

Open solution

Find the correct statements related to group 15 hydrides.

A. Reducing nature increases from $NH_3$ to $BiH_3$

B. Tendency to donate lone pair of electrons decreases from $NH_3$ to $BiH_3$

C. The stability of hydrides decreases from $NH_3$ to $BiH_3$

D. HEH bond angle decreases from $NH_3$ to $SbH_3 (E = \text{Elements of group 15})$

Choose the correct answer from the options given below :

(A) A and B only (B) B and C only (C) A, B, C and D (D) A, C and D Only

**Q2.** JEE Main 2026 (Online) 6th April Evening Shift

Open solution

Treatment of a gas ' $X$ ' with a freshly prepared ferrous sulphate solution gives a compound ' $Y$ ' as a brown ring. The compounds X and Y are:

(A) $NO$ and $[Fe(NO)]SO_4$

(B) $NO_2$ and $[Fe(NO_2)]SO_4$

(C) $N_2O$ and $[Fe(N_2O)]SO_4$

(D) $N_2O_4$ and $[Fe(N_2O_4)]SO_4$

**Q3.** JEE Main 2026 (Online) 6th April Morning Shift

Open solution

Given below are two statements :

**Statement I :** Aluminium is more electropositive than thallium as the standard electrode potential value of $E^\circ Al^{3+}/Al$ is negative and $E^\circ Tl^{3+}/Tl$ is positive.

**Statement II :** The sum of first three ionization enthalpies of boron is very high when compared to that of aluminium. Due to this reason boron forms covalent compounds only and aluminium forms $Al^{3+}$ ion.

In the light of the above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

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**Q4.** JEE Main 2026 (Online) 5th April Evening Shift

Open solution

Given below are two statements :

Statement I: Aluminium upon reaction with NaOH forms $[Al(OH)_6]^{3-}$ ion.

Statement II : The geometry of $ICl_4^-$, $ClO_3^-$ and $IBr_2^-$ is square planar, pyramidal and linear respectively.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

**Q5.** JEE Main 2026 (Online) 5th April Morning Shift

Open solution

Correct statements from the following are:

A. Nitrogen in oxidation states from +1 to +4 disproportionates in acid medium.

B. Nitrogen has the ability to form $d\pi$ - $p\pi$ multiple bonds with itself and other elements with small size and high electronegativity.

C. N-N single bond is stronger than P-P single bond.

D. Nitrogen has highest density in its group due to small size.

E. The maximum covalency of nitrogen is four since it has only four valence orbitals for bonding.

Choose the correct answer from the options given below:

(A) B, C and D Only (B) C, D and E Only (C) A, C and E Only (D) A and E Only

**Q6.** JEE Main 2026 (Online) 4th April Evening Shift

Open solution

The electronegativity of a group 13 element ' E ' is same as that of Ge (on Pauling scale and upto one decimal point). The CORRECT statements about $E^{3+}$ are

A. It can act as a reducing agent.

B. It can act as an oxidizing agent.

C. $E^{3+}$ is more stable than $E^+$.

D. The standard electrode potential value for $E^{3+}/E$ is positive.

Choose the correct answer from the options given below :

(A) A and C Only (B) B and C Only (C) B and D Only (D) A and D Only

**Q7.** JEE Main 2026 (Online) 4th April Morning Shift

Open solution

A monoatomic anion $(A^-)$ has 45 neutrons and 36 electrons. Atomic mass, group in the periodic table and physical state at room temperature of the element (A) respectively are

(A) 80, 17, liquid (B) 81, 16, solid (C) 80,16, gas (D) 81, 15, gas

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**Q8.** JEE Main 2026 (Online) 4th April Morning Shift

Open solution

Given below are two statements :

**Statement I :** The covalency of oxygen is generally two but it can exceed upto four. The oxidation state of oxygen in $SO_2$ is -2 and in $OF_2$ it is +2 .

**Statement II :** The anomalous behaviour of oxygen when compared to the other elements of group 16 is due to its small size and high electronegativity.

In the light of the above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

**Q9.** JEE Main 2026 (Online) 2nd April Evening Shift

Open solution

Given below are two statements :

**Statement I:** The second ionization enthalpy of B, Al and Ga is in the order of B > Al > Ga.

**Statement II:** The correct order in terms of first ionization enthalpy is Si < Ge < Pb < Sn.

In the light of the above statements, choose the **correct** answer from the options given below :

(A) Both **Statement I** and **Statement II** are true

(B) Both **Statement I** and **Statement II** are false

(C) **Statement I** is true but **Statement II** is false

(D) **Statement I** is false but **Statement II** is true

**Q10.** JEE Main 2026 (Online) 28th January Evening Shift

Open solution

Given below are two statements :

**Statement I :** The increasing order of boiling point of hydrogen halides is HCl < HBr < HI < HF.

**Statement II :** The increasing order of melting point of hydrogen halides is HCl < HBr < HF < HI.

In the light of the above statements, choose the **correct** answer from the options given below :

(A) **Statement I** is false but **Statement II** is true

(B) Both **Statement I** and **Statement II** are false

(C) **Statement I** is true but **Statement II** is false

(D) Both **Statement I** and **Statement II** are true

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**Q11.** JEE Main 2026 (Online) 28th January Morning Shift

Open solution

Regarding the hydrides of group 15 elements $EH_3(E = N, P, As, Sb)$, select the correct statement from the following :

A. The stability of hydrides decreases down the group.

B. The basicity of hydrides decreases down the group.

C. The reducing character increases down the group.

D. The boiling point increases down the group.

Choose the correct answer from the options given below :

**(A)** A, B & C only **(B)** A, B, C& D **(C)** A & D only **(D)** B & C only

**Q12.** JEE Main 2026 (Online) 24th January Evening Shift

Open solution

Choose the **INCORRECT** statement

**(A)** Carbon exhibits negative oxidation states along with +4 and +2 .

**(B)** Carbon cannot exceed its covalency more than four.

**(C)** Among the isotopes of carbon, $^{13}C$ is a radioactive isotope.

**(D)** $CO_2$ is the most acidic oxide among the dioxides of group of 14 elements.

**Q13.** JEE Main 2026 (Online) 23rd January Evening Shift

Open solution

It is noticed that $Pb^{2+}$ is more stable than $Pb^{4+}$ but $Sn^{2+}$ is less stable than $Sn^{4+}$. Observe the following reactions.

$PbO_2 + Pb \rightarrow 2PbO; \Delta_r G^\circ(1)$

$SnO_2 + Sn \rightarrow 2SnO; \Delta_r G^\circ(2)$

Identify the correct set from the following

**(A)** $\Delta_r G^\circ(1) > 0; \Delta_r G^\circ(2) < 0$

**(B)** $\Delta_r G^\circ(1) < 0; \Delta_r G^\circ(2) > 0$

**(C)** $\Delta_r G^\circ(1) < 0; \Delta_r G^\circ(2) < 0$

**(D)** $\Delta_r G^\circ(1) > 0; \Delta_r G^\circ(2) > 0$

**Q14.** JEE Main 2026 (Online) 23rd January Morning Shift

Open solution

The correct statements from the following are :

A. Ionic radii of trivalent cations of group 13 elements decreases down the group.

B. Electronegativity of group 13 elements decreases down the group.

C. Among the group 13 elements, Boron has highest first ionisation enthalpy.

D. The trichloride and triiodide of group 13 elements are covalent in nature.

Choose the correct answer from the options given below :

**(A)** B and D Only **(B)** C and D Only **(C)** A and D Only **(D)** A and C Only

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**Q15.** JEE Main 2026 (Online) 22nd January Evening Shift

Open solution

Given below are two statements :

**Statement I :** Elements ' $X$ ' and ' $Y$ ' are the most and least electronegative elements, respectively among $\mathrm{N}, \mathrm{As}, \mathrm{Sb}$ and P . The nature of the oxides $\mathrm{X_2O_3}$ and $\mathrm{Y_2O_3}$ is acidic and amphoteric, respectively.

**Statement II :** $\mathrm{BCl_3}$ is covalent in nature and gets hydrolysed in water. It produces $[\mathrm{B(OH)_4}]^-$ and $[\mathrm{B(H_2O)_6}]^{3+}$ in aqueous medium.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is false but Statement II is true

(B) Statement I is true but Statement II is false

(C) Both Statement I and Statement II are false

(D) Both Statement I and Statement II are true

**Q16.** JEE Main 2026 (Online) 22nd January Morning Shift

Open solution

Given below are two statements :

**Statement I :** The halogen that makes longest bond with hydrogen in HX, has the smallest covalent radius in its group.

**Statement II :** A group 15 element's hydride $\mathrm{EH_3}$ has the lowest boiling point among corresponding hydrides of other group 15 elements. The maximum covalency of that element E is 4 .

In the light of the above statements, choose the correct answer from the options given below

(A) Statement I is true but Statement II is false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are true

(D) Both Statement I and Statement II are false

**Q17.** JEE Main 2026 (Online) 22nd January Morning Shift

Open solution

A 'p'-block element $(\mathrm{E})$ and hydrogen form a binary cation $(\mathrm{EH_x})^+$, while $\mathrm{EH_3}$ on treatment with $\mathrm{K_2HgI_4}$ in alkaline medium gives a precipitate of basic mercury(II)amido-iodine. Given below are first ionisation enthalpy values $\left(\mathrm{kJmol^{-1}}\right)$ for first element each from group 13, 14, 15 and 16. Identify the correct first ionisation enthalpy value for element E.

(A) 1402 (B) 1086 (C) 801 (D) 1312

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**Q18.** JEE Main 2026 (Online) 21st January Morning Shift

Open solution

Consider the following reactions.

$$\underset{\text{(Hot solution)}}{PbCl_2} + K_2CrO_4 \rightarrow A + 2KCl$$

$$A + NaOH \rightleftharpoons B + Na_2CrO_4$$

$$PbSO_4 + 4CH_3COONH_4 \rightarrow (NH_4)_2SO_4 + X$$

In the above reactions, $A, B$ and X are respectively.

**(A)** $Na_2[Pb(OH)_2], PbCrO_4$ and $[Pb(NH_3)_4]SO_4$

**(B)** $Na_2[Pb(OH)_2], PbCrO_4$ and $(NH_4)_2[Pb(CH_3COO)_4]$

**(C)** $PbCrO_4, Na_2[Pb(OH)_4]$ and $(NH_4)_2[Pb(CH_3COO)_4]$

**(D)** $PbCrO_4, Na_2[Pb(OH)_4]$ and $[Pb(NH_3)_4]SO_4$

**Q19.** JEE Main 2026 (Online) 21st January Morning Shift

Open solution

Given below are two statements :

**Statement I :** The number of pairs among $[SiO_2, CO_2], [SnO, SnO_2], [PbO, PbO_2]$ and $[GeO, GeO_2]$, which contain oxides that are both amphoteric is 2 .

**Statement II :** $BF_3$ is an electron deficient molecule, can act as a Lewis acid, forms adduct with $NH_3$ and has a trigonal planar geometry.

In the light of the above statements, choose the correct answer from the options given below :

**(A)** Both Statement I and Statement II are false

**(B)** Statement I is false but Statement II is true

**(C)** Both Statement I and Statement II are true

**(D)** Statement I is true but Statement II is false

**Q20.** JEE Main 2025 (Online) 7th April Evening Shift

Open solution

The correct statements from the following are :

(A) $Tl^{3+}$ is a powerful oxidising agent

(B) $Al^{3+}$ does not get reduced easily

(C) Both $Al^{3+}$ and $Tl^{3+}$ are very stable in solution

(D) $Tl^{1+}$ is more stable than $Tl^{3+}$

(E) $Al^{3+}$ and $Tl^{1+}$ are highly stable

Choose the **correct** answer from the options given below :

**(A)** (A), (B), (D) and (E) only

**(B)** (A), (C) and (D) only

**(C)** (A), (B), (C) and (E)

**(D)** (B), (D) and (E) only

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**Q21.** JEE Main 2025 (Online) 4th April Evening Shift

Open solution

Given below are two statements :

Statement (I) : The first ionisation enthalpy of group 14 elements is higher than the corresponding elements of group 13.

Statement (II) : Melting points and boiling points of group 13 elements are in general much higher than those of corresponding elements of group 14.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is correct but Statement II is incorrect

(B) Both Statement I and Statement II are incorrect

(C) Statement I is incorrect but Statement II is correct

(D) Both Statement I and Statement II are correct

**Q22.** JEE Main 2025 (Online) 4th April Morning Shift

Open solution

Given below are two statements:

Statement I: Nitrogen forms oxides with +1 to +5 oxidation states due to the formation of $p\pi - p\pi$ bond with oxygen.

Statement II: Nitrogen does not form halides with +5 oxidation state due to the absence of d-orbital in it.

In the light of given statements, choose the correct answer from the options given below.

(A) Both Statement I and Statement II are true

(B) Statement I is true but Statement II is false

(C) Statement I is false but Statement II is true

(D) Both Statement I and Statement II are False

**Q23.** JEE Main 2025 (Online) 4th April Morning Shift

Open solution

Given below are the pairs of group 13 elements showing their relation in terms of atomic radius. $(B < Al), (Al < Ga), (Ga < In)$ and $(In < Tl)$ Identify the elements present in the incorrect pair and in that pair find out the element (X) that has higher ionic radius $(M^{3+})$ than the other one. The atomic number of the element $(X)$ is

(A) 49 (B) 81 (C) 31 (D) 13

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**Q24.** JEE Main 2025 (Online) 3rd April Morning Shift

Open solution

Given below are two statements :

**Statement I :** The N - N single bond is weaker and longer than that of P - P single bond.

**Statement II :** Compounds of group 15 elements in +3 oxidation states readily undergo disproportionation reactions.

In the light of the above statements, choose the correct answer from the options given below

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are true

(C) Both Statement I and Statement II are false

(D) Statement I is true but Statement II is false

**Q25.** JEE Main 2025 (Online) 2nd April Evening Shift

Open solution

The nature of oxide $(TeO_2)$ and hydride $(TeH_2)$ formed by Te , respectively are :

(A) Reducing and basic

(B) Reducing and acidic

(C) Oxidising and acidic

(D) Oxidising and basic

**Q26.** JEE Main 2025 (Online) 24th January Morning Shift

Open solution

The large difference between the melting and boiling points of oxygen and sulphur may be explained on the basis of

(A) Atomicity

(B) Electron gain enthalpy

(C) Atomic size

(D) Electronegativity

**Q27.** JEE Main 2025 (Online) 23rd January Evening Shift

Open solution

Given below are the atomic numbers of some group 14 elements. The atomic number of the element with lowest melting point is :

(A) 14

(B) 50

(C) 6

(D) 82

**Q28.** JEE Main 2025 (Online) 23rd January Morning Shift

Open solution

The incorrect statement among the following is

(A) $PH_3$ shows lower proton affinity than $NH_3$.

(B) $SO_2$ can act as an oxidizing agent, but not as a reducing agent.

(C) $NO_2$ can dimerise easily.

(D) $PF_3$ exists but $NF_5$ does not.

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**Q29.** JEE Main 2025 (Online) 22nd January Evening Shift

The maximum covalency of a non-metallic group 15 element ' E ' with weakest $E - E$ bond is :

**(A)** 5 **(B)** 3 **(C)** 4 **(D)** 6

Open solution

**Q30.** JEE Main 2024 (Online) 9th April Morning Shift

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Both rhombic and monoclinic sulphur exist as $S_8$ while oxygen exists as $O_2$.

Reason (R) : Oxygen forms $p\pi - p\pi$ multiple bonds with itself and other elements having small size and high electronegativity like $C, N$, which is not possible for sulphur.

In the light of the above statements, choose the most appropriate answer from the options given below :

**(A)** Both (A) and (R) are correct but (R) is not the correct explanation of (A)

**(B)** (A) is correct but (R) is not correct

**(C)** (A) is not correct but (R) is correct

**(D)** Both $(\mathbf{A})$ and $(\mathbf{R})$ are correct and $(\mathbf{R})$ is the correct explanation of $(\mathbf{A})$

Open solution

**Q31.** JEE Main 2024 (Online) 9th April Morning Shift

On reaction of Lead Sulphide with dilute nitric acid which of the following is not formed?

**(A)** Sulphur **(B)** Nitric oxide **(C)** Lead nitrate **(D)** Nitrous oxide

Open solution

**Q32.** JEE Main 2024 (Online) 8th April Evening Shift

Identify the correct statements about p-block elements and their compounds.

(A) Non metals have higher electronegativity than metals.

(B) Non metals have lower ionisation enthalpy than metals.

(C) Compounds formed between highly reactive nonmetals and highly reactive metals are generally ionic.

(D) The non-metal oxides are generally basic in nature.

(E) The metal oxides are generally acidic or neutral in nature.

Choose the correct answer from the options given below :

**(A)** (B) and (D) only **(B)** (B) and (E) only **(C)** (D) and (E) only **(D)** (A) and (C) only

Open solution

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**Q33.** JEE Main 2024 (Online) 8th April Evening Shift

Open solution

Identify the incorrect statements about group 15 elements :

(A) Dinitrogen is a diatomic gas which acts like an inert gas at room temperature.

(B) The common oxidation states of these elements are $-3, +3$ and +5.

(C) Nitrogen has unique ability to form $p\pi - p\pi$ multiple bonds.

(D) The stability of +5 oxidation states increases down the group.

(E) Nitrogen shows a maximum covalency of 6.

Choose the correct answer from the options given below :

**(A)** (A), (C), (E) only

**(B)** (D) and (E) only

**(C)** (A), (B), (D) only

**(D)** (B), (D), (E) only

**Q34.** JEE Main 2024 (Online) 8th April Evening Shift

Open solution

In qualitative test for identification of presence of phosphorus, the compound is heated with an oxidising agent. Which is further treated with nitric acid and ammonium molybdate respectively. The yellow coloured precipitate obtained is :

**(A)** $Na_3PO_4 \cdot 12MoO_3$

**(B)** $(NH_4)_3PO_4 \cdot 12(NH_4)_2MoO_4$

**(C)** $MoPO_4 \cdot 21NH_4NO_3$

**(D)** $(NH_4)_3PO_4 \cdot 12MoO_3$

**Q35.** JEE Main 2024 (Online) 8th April Morning Shift

Open solution

Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The stability order of +1 oxidation state of Ga, In and Tl is Ga < In < Tl.

Reason R: The inert pair effect stabilizes the lower oxidation state down the group.

In the light of the above statements, choose the correct answer from the options given below:

**(A)** A is true but R is false.

**(B)** Both A and R are true but R is NOT the correct explanation of A.

**(C)** A is false but R is true.

**(D)** Both A and R are true and R is the correct explanation of A.

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**Q36.** JEE Main 2024 (Online) 8th April Morning Shift

Open solution

Among the following halogens

$F_2, Cl_2, Br_2$ and $I_2$

Which can undergo disproportionation reactions?

**(A)** $F_2$ and $Cl_2$

**(B)** $Cl_2, Br_2$ and $I_2$

**(C)** Only $I_2$

**(D)** $F_2, Cl_2$ and $Br_2$

**Q37.** JEE Main 2024 (Online) 6th April Evening Shift

Open solution

Evaluate the following statements related to group 14 elements for their correctness.

(A) Covalent radius decreases down the group from C to Pb in a regular manner.

(B) Electronegativity decreases from C to Pb down the group gradually.

(C) Maximum covalency of C is 4 whereas other elements can expand their covalency due to presence of d orbitals.

(D) Heavier elements do not form $p\pi - p\pi$ bonds.

(E) Carbon can exhibit negative oxidation states.

Choose the correct answer from the options given below :

**(A)** (A), (B) and (C) Only

**(B)** (A) and (B) Only

**(C)** (C), (D) and (E) Only

**(D)** (C) and (D) Only

**Q38.** JEE Main 2024 (Online) 6th April Morning Shift

Open solution

Given below are two statements :

Statement I : Gallium is used in the manufacturing of thermometers.

Statement II : A thermometer containing gallium is useful for measuring the freezing point $(256\ K)$ of brine solution.

In the light of the above statements, choose the correct answer from the options given below :

**(A)** Both Statement I and Statement II are true

**(B)** Statement I is true but Statement II is false

**(C)** Statement I is false but Statement II is true

**(D)** Both Statement I and Statement II are false

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**Q39.** JEE Main 2024 (Online) 5th April Evening Shift

Open solution

The correct statements from the following are :

(A) The decreasing order of atomic radii of group 13 elements is $\mathrm{Tl} > \mathrm{In} > \mathrm{Ga} > \mathrm{Al} > \mathrm{B}$.

(B) Down the group 13 electronegativity decreases from top to bottom.

(C) $\mathrm{Al}$ dissolves in dil. $\mathrm{HCl}$ and liberates $\mathrm{H_2}$ but conc. $\mathrm{HNO_3}$ renders $\mathrm{Al}$ passive by forming a protective oxide layer on the surface.

(D) All elements of group 13 exhibits highly stable +1 oxidation state.

(E) Hybridisation of $\mathrm{Al}$ in $[\mathrm{Al(H_2O)_6}]^{3+}$ ion is $\mathrm{sp^3d^2}$.

Choose the correct answer from the options given below :

**(A)** (A), (B), (C) and (E) only

**(B)** (C) and (E) only

**(C)** (A), (C) and (E) only

**(D)** (A) and (C) only

**Q40.** JEE Main 2024 (Online) 5th April Morning Shift

Open solution

Given below are two statements:

Statement I: In group 13, the stability of +1 oxidation state increases down the group.

Statement II : The atomic size of gallium is greater than that of aluminium.

In the light of the above statements, choose the most appropriate answer from the options given below :

**(A)** Statement I is incorrect but Statement II is correct

**(B)** Statement I is correct but Statement II is incorrect

**(C)** Both Statement I and Statement II are correct

**(D)** Both Statement I and Statement II are incorrect

**Q41.** JEE Main 2024 (Online) 5th April Morning Shift

Open solution

The number of neutrons present in the more abundant isotope of boron is '$x$'. Amorphous boron upon heating with air forms a product, in which the oxidation state of boron is '$y$'. The value of $x + y$ is _____ .

**(A)** 9 **(B)** 6 **(C)** 4 **(D)** 3

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**Q42.** JEE Main 2024 (Online) 1st February Evening Shift

Open solution

Given below are two statements :

**Statement (I)** : $SiO_2$ and $GeO_2$ are acidic while $SnO$ and $PbO$ are amphoteric in nature.

**Statement (II)** : Allotropic forms of carbon are due to property of catenation and $p\pi - d\pi$ bond formation.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are false

(C) Both Statement I and Statement II are true

(D) Statement I is true but Statement II is false

**Q43.** JEE Main 2024 (Online) 1st February Evening Shift

Open solution

The strongest reducing agent among the following is :

(A) $SbH_3$ (B) $NH_3$ (C) $BiH_3$ (D) $PH_3$

**Q44.** JEE Main 2024 (Online) 31st January Evening Shift

Open solution

Given below are two statements:

**Statement I** : Group 13 trivalent halides get easily hydrolyzed by water due to their covalent nature.

**Statement II** : $AlCl_3$ upon hydrolysis in acidified aqueous solution forms octahedral $[Al(H_2O)_6]^{3+}$ ion.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is false but statement II is true

(B) Both statement I and statement II are true

(C) Both statement I and statement II are false

(D) Statement I is true but statement II is false

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**Q45.** JEE Main 2024 (Online) 31st January Evening Shift

Open solution

Given below are two statements:

**Statement I** : $S_8$ solid undergoes disproportionation reaction under alkaline conditions to form $S^{2-}$ and $S_2O_3^{2-}$.

**Statement II** : $ClO_4^-$ can undergo disproportionation reaction under acidic condition.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(A) Statement I is correct but statement II is incorrect

(B) Both statement I and statement II are incorrect

(C) Statement I is incorrect but statement II is correct

(D) Both statement I and statement II are correct

**Q46.** JEE Main 2024 (Online) 31st January Evening Shift

Open solution

Choose the correct statements from the following

A. All group 16 elements form oxides of general formula $EO_2$ and $EO_3$, where $E = S, Se, Te$ and $Po$. Both the types of oxides are acidic in nature.

B. $TeO_2$ is an oxidising agent while $SO_2$ is reducing in nature.

C. The reducing property decreases from $H_2S$ to $H_2Te$ down the group.

D. The ozone molecule contains five lone pairs of electrons.

Choose the correct answer from the options given below:

(A) A and B only (B) C and D only (C) A and D only (D) B and C only

**Q47.** JEE Main 2024 (Online) 31st January Morning Shift

Open solution

Consider the oxides of group 14 elements $SiO_2, GeO_2, SnO_2, PbO_2, CO$ and $GeO$. The amphoteric oxides are

(A) $SnO_2, CO$ (B) $SiO_2, GeO_2$ (C) $SnO_2, PbO_2$ (D) $GeO, GeO_2$

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**Q48.** JEE Main 2024 (Online) 31st January Morning Shift

Open solution

Given below are two statements:

Statement - I: Noble gases have very high boiling points.

Statement - II: Noble gases are monoatomic gases. They are held together by strong dispersion forces. Because of this they are liquefied at very low temperature. Hence, they have very high boiling points.

In the light of the above statements, choose the correct answer from the options given below:

(A) Both Statement I and Statement II are false.

(B) Statement I is true but Statement II is false.

(C) Statement I is false but Statement II is true.

(D) Both Statement I and Statement II are true.

**Q49.** JEE Main 2024 (Online) 30th January Evening Shift

Open solution

Choose the correct statements about the hydrides of group 15 elements.

A. The stability of the hydrides decreases in the order $NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$.

B. The reducing ability of the hydride increases in the order $NH_3 < PH_3 < AsH_3 < SbH_3 < BiH_3$.

C. Among the hydrides, $NH_3$ is strong reducing agent while $BiH_3$ is mild reducing agent.

D. The basicity of the hydrides increases in the order $NH_3 < PH_3 < AsH_3 < SbH_3 < BiH_3$.

Choose the most appropriate from the options given below :

(A) C and D only (B) A and D only (C) A and B only (D) B and C only

**Q50.** JEE Main 2024 (Online) 29th January Evening Shift

Open solution

On passing a gas, 'X', through Nessler's reagent, a brown precipitate is obtained. The gas 'X' is

(A) $Cl_2$ (B) $CO_2$ (C) $NH_3$ (D) $H_2S$

**Q51.** JEE Main 2024 (Online) 29th January Morning Shift

Open solution

Identify the incorrect pair from the following :

(A) Fluoroapatite $-3Ca_3(PO_4)_2 \cdot CaF_2$

(B) Carnallite $-KCl \cdot MgCl_2 \cdot 6H_2O$

(C) Cryolite $-Na_3AlF_6$

(D) Fluorspar $-BF_3$

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**Q52.** JEE Main 2024 (Online) 27th January Evening Shift

Open solution

Given below are two statements :

Statement (I) : Oxygen being the first member of group 16 exhibits only -2 oxidation state.

Statement (II) : Down the group 16 stability of +4 oxidation state decreases and +6 oxidation state increases.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is correct but Statement II is incorrect

(B) Both Statement I and Statement II are incorrect

(C) Statement I is incorrect but Statement II is correct

(D) Both Statement I and Statement II are correct

**Q53.** JEE Main 2024 (Online) 27th January Morning Shift

Open solution

Element not showing variable oxidation state is :

(A) Bromine (B) Iodine (C) Chlorine (D) Fluorine

**Q54.** JEE Main 2024 (Online) 27th January Morning Shift

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Melting point of Boron (2453 K) is unusually high in group 13 elements.

Reason (R) : Solid Boron has very strong crystalline lattice.

In the light of the above statements, choose the most appropriate answer from the options given below ;

(A) Both (A) and (R) are correct but (R) Is not the correct explanation of (A)

(B) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(C) (A) is true but (R) is false

(D) (A) is false but (R) is true

**Q55.** JEE Main 2023 (Online) 15th April Morning Shift

Open solution

The number of $P-O-P$ bonds in $H_4P_2O_7$, $(HPO_3)_3$ and $P_4O_{10}$ are respectively :

(A) $1,3,6$ (B) $1,2,4$ (C) $0,3,4$ (D) $0,3,6$

**Q56.** JEE Main 2023 (Online) 13th April Evening Shift

Open solution

The covalency and oxidation state respectively of boron in $[BF_4]^-$, are :

(A) 3 and 4 (B) 4 and 3 (C) 4 and 4 (D) 3 and 5

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**Q57.** JEE Main 2023 (Online) 12th April Morning Shift

Open solution

Given below are two statements :

Statement I : Boron is extremely hard indicating its high lattice energy.

Statement II : Boron has highest melting and boiling point compared to its other group members.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are incorrect

(B) Statement I is incorrect but Statement II is correct

(C) Statement I is correct but Statement II is incorrect

(D) Both statement I and Statement II are correct

**Q58.** JEE Main 2023 (Online) 12th April Morning Shift

Open solution

Given below are two statements :

Statement I : $SbCl_5$ is more covalent than $SbCl_3$

Statement II: The higher oxides of halogens also tend to be more stable than the lower ones.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both statement I and Statement II are correct

(B) Both Statement I and Statement II are incorrect

(C) Statement I is correct but Statement II is incorrect

(D) Statement I is incorrect but Statement II is correct

**Q59.** JEE Main 2023 (Online) 11th April Morning Shift

Open solution

For compound having the formula $GaAlCl_4$, the correct option from the following is :

(A) $Cl$ forms bond with both $Al$ and $Ga$ in $GaAlCl_4$

(B) Oxidation state of $Ga$ in the salt $GaAlCl_4$ is +3 .

(C) $Ga$ is coordinated with $Cl$ in $GaAlCl_4$

(D) $Ga$ is more electronegative than $Al$ and is present as a cationic part of the salt $GaAlCl_4$

**Q60.** JEE Main 2023 (Online) 24th January Evening Shift

Open solution

$K_2Cr_2O_7$ paper acidified with dilute $H_2SO_4$ turns green when exposed to :

(A) Sulphur dioxide

(B) Sulphur trioxide

(C) Hydrogen sulphide

(D) Carbon dioxide

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**Q61.** JEE Main 2022 (Online) 29th July Evening Shift

Open solution

Given below are two statements.

Statement I : Stannane is an example of a molecular hydride.

Statement II : Stannane is a planar molecule.

In the light of the above statement, choose the most appropriate answer from the options given below.

**(A)** Both Statement I and Statement II are true.

**(B)** Both Statement I and Statement II are false.

**(C)** Statement I is true but Statement II is false.

**(D)** Statement I is false but Statement II is true.

**Q62.** JEE Main 2022 (Online) 28th July Evening Shift

Open solution

Dinitrogen and dioxygen, the main constituents of air do not react with each other in atmosphere to form oxides of nitrogen because :

**(A)** $N_2$ is unreactive in the condition of atmosphere.

**(B)** Oxides of nitrogen are unstable.

**(C)** Reaction between them can occur in the presence of a catalyst.

**(D)** The reaction is endothermic and require very high temperature.

**Q63.** JEE Main 2022 (Online) 26th July Morning Shift

Open solution

Match List - I with List - II.

| | **List I (Processes / Reactions)** | | **List II (Catalyst)** |
|---|---|---|---|
| (A) | $2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$ | (I) | Fe(s) |
| (B) | $4NH_3(g) + 5O_2(g) \rightarrow 4NO(g) + 6H_2O(g)$ | (II) | Pt(s) – Rh(s) |
| (C) | $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$ | (III) | $V_2O_5$ |
| (D) | Vegetable oil(l) + $H_2 \rightarrow$ Vegetable ghee(s) | (IV) | Ni(s) |

Choose the correct answer from the options given below :

**(A)** $(A) - (III), (B) - (I), (C) - (II), (D) - (IV)$

**(B)** $(A) - (III), (B) - (II), (C) - (I), (D) - (IV)$

**(C)** $(A) - (IV), (B) - (III), (C) - (I), (D) - (II)$

**(D)** $(A) - (IV), (B) - (II), (C) - (III), (D) - (I)$

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**Q64.** JEE Main 2022 (Online) 25th July Evening Shift

Open solution

Match List I with List II:

| | List I | | List II |
|---|---|---|---|
| (A) | Sulphate | (I) | Pesticide |
| (B) | Fluoride | (II) | Bending of bones |
| (C) | Nicotine | (III) | Laxative effect |
| (D) | Sodium arsenite | (IV) | Herbicide |

Choose the correct answer from the options given below :

**(A)** A-II, B-III, C-IV, D-I

**(B)** A-IV, B-III, C-II, D-I

**(C)** A-III, B-II, C-I, D-IV

**(D)** A-III, B-II, C-IV, D-I

**Q65.** JEE Main 2022 (Online) 30th June Morning Shift

Open solution

A $\xrightarrow{573\ K}$ Red phosphorus $\xrightarrow[under\ pressure]{heat\ ;\ 803\ K}$ B

Red phosphorus is obtained by heating "A" at 573 K, and can be converted to "B" by heating at 803 K under pressure.

A and B, respectively, are

**(A)** $\beta$-black phosphorus and white phosphorus.

**(B)** white phosphorus and $\beta$-black phosphorus.

**(C)** $\alpha$-black phosphorus and white phosphorus.

**(D)** white phosphorus and $\alpha$-black phosphorus.

**Q66.** JEE Main 2022 (Online) 29th June Evening Shift

Open solution

Sulphur dioxide is one of the components of polluted air. $SO_2$ is also a major contributor to acid rain. The correct and complete reaction to represent acid rain caused by $SO_2$ is :

**(A)** $2SO_2 + O_2 \rightarrow 2SO_3$

**(B)** $SO_2 + O_3 \rightarrow SO_3 + O_2$

**(C)** $SO_2 + H_2O_2 \rightarrow H_2SO_4$

**(D)** $2SO_2 + O_2 + 2H_2O \rightarrow 2H_2SO_4$

**Q67.** JEE Main 2022 (Online) 28th June Morning Shift

Open solution

Nitrogen gas is obtained by thermal decomposition of :

**(A)** $Ba(NO_3)_2$

**(B)** $Ba(N_3)_2$

**(C)** $NaNO_2$

**(D)** $NaNO_3$

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**Q68.** JEE Main 2022 (Online) 27th June Evening Shift

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : Fluorine forms one oxoacid.

Reason R : Fluorine has smallest size amongst all halogens and is highly electronegative.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both A and R are correct and R is the correct explanation of A.

(B) Both A and R are correct but R is NOT the correct explanation of A.

(C) A is correct but R is not correct.

(D) A is not correct but R is correct.

Open solution

**Q69.** JEE Main 2022 (Online) 26th June Evening Shift

Which of the following elements is considered as a metalloid?

(A) Sc (B) Pb (C) Bi (D) Te

Open solution

**Q70.** JEE Main 2022 (Online) 26th June Evening Shift

The oxide which contains an odd electron at the nitrogen atom is :

(A) $N_2O$ (B) $NO_2$ (C) $N_2O_3$ (D) $N_2O_5$

Open solution

**Q71.** JEE Main 2022 (Online) 26th June Morning Shift

Choose the correct stability order of group 13 elements in their + 1 oxidation state :

(A) Al < Ga < In < Tl (B) Tl < In < Ga < Al

(C) Al < Ga < Tl < In (D) Al < Tl < Ga < In

Open solution

**Q72.** JEE Main 2022 (Online) 24th June Evening Shift

$PCl_5$ is well known, but $NCl_5$ is not. because,

(A) nitrogen is less reactive than phosphorus.

(B) nitrogen doesn't have d-orbitals in its valence shell.

(C) catenation tendency is weaker in nitrogen than phosphorus.

(D) size of phosphorus is larger than nitrogen.

Open solution

**Q73.** JEE Main 2022 (Online) 24th June Morning Shift

The most stable trihalide of nitrogen is :

(A) $NF_3$ (B) $NCl_3$ (C) $NBr_3$ (D) $NI_3$

Open solution

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**Q74.** JEE Main 2021 (Online) 27th August Morning Shift

In polythionic acid, $H_2S_xO_6$ (x = 3 to 5) the oxidation state(s) of sulphur is/are :

(A) + 5 only (B) + 6 only (C) + 3 and + 5 only (D) 0 and + 5 only

Open solution

**Q75.** JEE Main 2021 (Online) 27th July Evening Shift

Match List - I with List - II :

| | List - I (compound) | | List - II (effect/affected species) |
|---|---|---|---|
| (a) | Carbon monoxide | (i) | Carcinogenic |
| (b) | Sulphur dioxide | (ii) | Metabolized by pyrus plants |
| (c) | Polychlorinated biphenyls | (iii) | haemoglobin |
| (d) | Oxides of Nitrogen | (iv) | Stiffness of flower buds |

Choose the correct answer from the options given below :

(A) (a) - (iii), (b) - (iv), (c) - (i), (d) - (ii)

(B) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

(C) (a) - (i), (b) - (ii), (c) - (iii), (d) - (iv)

(D) (a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)

Open solution

**Q76.** JEE Main 2021 (Online) 25th July Morning Shift

Which one of the following compounds of Group-14 elements is not known?

(A) $[GeCl_6]^{2-}$ (B) $[Sn(OH)_6]^{2-}$ (C) $[SiCl_6]^{2-}$ (D) $[SiF_6]^{2-}$

Open solution

**Q77.** JEE Main 2021 (Online) 22th July Evening Shift

Which one of the following group-15 hydride is the strongest reducing agent?

(A) $AsH_3$ (B) $BiH_3$ (C) $PH_3$ (D) $SbH_3$

Open solution

**Q78.** JEE Main 2021 (Online) 20th July Morning Shift

The species given below that does NOT show disproportionation reaction is :

(A) $BrO_4^-$ (B) $BrO^-$ (C) $BrO_2^-$ (D) $BrO_3^-$

Open solution

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**Q79.** JEE Main 2021 (Online) 18th March Morning Shift

Open solution

Match List - I with List - II

| | List - I (process) | | List - II (catalyst) |
|---|---|---|---|
| (a) | Deacon's process | (i) | ZSM-5 |
| (b) | Contact process | (ii) | $CuCl_2$ |
| (c) | Cracking of hydrocarbons | (iii) | Particles 'Ni' |
| (d) | Hydrogenation of vegetable oils | (iv) | $V_2O_5$ |

Choose the most appropriate answer from the options given below :

**(A)** (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

**(B)** (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)

**(C)** (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

**(D)** (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)

**Q80.** JEE Main 2021 (Online) 16th March Morning Shift

Open solution

An element of group 15, which is a metal and forms a hydride with strongest reducing power among group 15 hydrides. The element is :

**(A)** As **(B)** Sb **(C)** Bi **(D)** P

**Q81.** JEE Main 2021 (Online) 26th February Evening Shift

Open solution

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : In $TlI_3$, isomorphous to $CsI_3$, the metal is present in +1 oxidation state.

Reason R : Tl metal has fourteen f electrons in its electronic configuration.

In the light of the above statements, choose the most appropriate answer from the options given below :

**(A)** A is correct but R is not correct

**(B)** A is not correct but R is correct

**(C)** Both A and R are correct but R is NOT the correct explanation of A

**(D)** Both A and R are correct and R is the correct explanation of A

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**Q82.** JEE Main 2021 (Online) 26th February Morning Shift

Open solution

Find A, B and C in the following reactions :

$$NH_3 + A + CO_2 \rightarrow (NH_4)_2CO_3$$

$$(NH_4)_2CO_3 + H_2O + B \rightarrow NH_4HCO_3$$

$$NH_4HCO_3 + NaCl \rightarrow NH_4Cl + C$$

(A) $A - O_2; B - CO_2; C - Na_2CO_3$

(B) $A - H_2O; B - O_2; C - NaHCO_3$

(C) $A - H_2O; B - O_2; C - Na_2CO_3$

(D) $A - H_2O; B - CO_2; C - NaHCO_3$

**Q83.** JEE Main 2020 (Online) 6th September Morning Slot

Open solution

The correct statement with respect to dinitrogen is :

(A) $N_2$ is paramagnetic in nature

(B) it can be used as an inert diluent for reactive chemicals

(C) it can combine with dioxygen at 25°C

(D) liquid dinitrogen is not used in cryosurgery

**Q84.** JEE Main 2020 (Online) 9th January Evening Slot

Open solution

The reaction of $H_3N_3B_3Cl_3$ (A) with $LiBH_4$ in tetrahydrofuran gives inorganic benzene (B). Further, the reaction of (A) with (C) leads to $H_3N_3B_3(Me)_3$. Compounds (B) and (C) respectively, are :

(A) Borazine and MeBr

(B) Diborane and MeMgBr

(C) Boron nitride and MeBr

(D) Borazine and MeMgBr

**Q85.** JEE Main 2020 (Online) 8th January Evening Slot

Open solution

A metal (A) on heating in nitrogen gas gives compound B. B on treatment with $H_2O$ gives a colourless gas which when passed through $CuSO_4$ solution gives a dark blue-violet coloured solution. A and B respectively, are :

(A) Na and $NaNO_3$

(B) Na and $Na_3N$

(C) Mg and $Mg_3N_2$

(D) Mg and $Mg(NO_3)_2$

**Q86.** JEE Main 2019 (Online) 10th April Evening Slot

Open solution

The noble gas that does not occur in the atmosphere is :

(A) Ne

(B) He

(C) Kr

(D) Rn

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**Q87.** JEE Main 2019 (Online) 10th April Morning Slot

The oxoacid of sulphur that does not contain bond between sulphur atoms is

(A) $H_2S_2O_7$ (B) $H_2S_2O_3$ (C) $H_2S_4O_6$ (D) $H_2S_2O_4$

Open solution

**Q88.** JEE Main 2019 (Online) 9th April Evening Slot

The correct statements among I to III regarding group 13 element oxides are,
(I) Boron trioxide is acidic.
(II) Oxides of aluminium and gallium are amphoteric.
(III) Oxides of indium and thallium are basic.

(A) (II) and (III) only (B) (I), (II) and (III)
(C) (I) and (III) only (D) (I) and (II) only

Open solution

**Q89.** JEE Main 2019 (Online) 12th January Evening Slot

The element that does NOT show catenation is :

(A) Ge (B) Si (C) Sn (D) Pb

Open solution

**Q90.** JEE Main 2019 (Online) 12th January Evening Slot

The element that shows greater ability to form $p\pi$-$p\pi$ multiple bonds, is :

(A) Si (B) Ge (C) C (D) Sn

Open solution

**Q91.** JEE Main 2019 (Online) 10th January Evening Slot

The pair that contains two P – H bonds in each of the oxoacids is

(A) $H_3PO_2$ and $H_4P_2O_5$ (B) $H_4P_2O_5$ and $H_4P_2O_6$
(C) $H_4P_2O_5$ and $H_3PO_3$ (D) $H_3PO_3$ and $H_3PO_2$

Open solution

**Q92.** JEE Main 2019 (Online) 9th January Morning Slot

Aluminium is usually found in +3 oxidation state. In contrast, thallium exists in + 1 and + 3 oxidation states. This is due to :

(A) inert pair effect (B) diagonal relationship
(C) lattice structure (D) lanthanoid contraction

Open solution

**Q93.** JEE Main 2019 (Online) 9th January Morning Slot

The one that is extensively used as a piezoelectric material is :

(A) tridymite (B) amorphous silica (C) quartz (D) mica

Open solution

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**Q94.** JEE Main 2018 (Offline)

The compound that does not produce nitrogen gas by the thermal decomposition is :

(A) $(NH_4)_2SO_4$ (B) $Ba(N_3)_2$ (C) $(NH_4)_2Cr_2O_7$ (D) $NH_4NO_2$

Open solution

**Q95.** JEE Main 2018 (Online) 15th April Evening Slot

In $KO_2$, the nature of oxygen species and the oxidation state of oxygen atom are, respectively :

(A) Oxide and – 2 (B) Superoxide and – 1/2

(C) Peroxide and – 1/2 (D) Superoxide and – 1

Open solution

**Q96.** JEE Main 2018 (Online) 15th April Morning Slot

In graphite and diamond, the percentage of p-characters of the hybrid orbitals in hybridisation are respectively :

(A) 33 and 25 (B) 33 and 75 (C) 50 and 75 (D) 67 and 75

Open solution

**Q97.** JEE Main 2017 (Online) 8th April Morning Slot

A metal 'M' reacts with nitrogen gas to afford '$M_3N$'. '$M_3N$' on heating at high temperature gives back 'M' and on reaction with water produces a gas 'B'. Gas 'B' reacts with aqueous solution of $CuSO_4$ to form a deep blue compound. 'M' and 'B' respectively are :

(A) Li and $NH_3$ (B) Ba and $N_2$ (C) Na and $NH_3$ (D) Al and $N_2$

Open solution

**Q98.** JEE Main 2016 (Online) 9th April Morning Slot

Which intermolecular force is most responsible in allowing xenon gas to liquefy?

(A) Dipole - dipole (B) Ion - dipole

(C) Instantaneous dipole - induced dipole (D) Ionic

Open solution

**Q99.** JEE Main 2016 (Online) 9th April Morning Slot

The non-metal that does **not** exhibit positive oxidation state is :

(A) Oxygen (B) Iodine (C) Chlorine (D) Fluorine

Open solution

**Q100.** JEE Main 2016 (Offline)

The reaction of zinc with dilute and concentrated nitric acid, respectively, produces :

(A) $NO_2$ and NO (B) NO and $N_2O$ (C) $NO_2$ and $N_2O$ (D) $N_2O$ and $NO_2$

Open solution

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**Q101.** JEE Main 2015 (Offline)

**Assertion :** Nitrogen and Oxygen are the main components in the atmosphere but these do not react to form oxides of nitrogen.

Open solution

**Reason :** The reaction between nitrogen and oxygen requires high temperature.

(A) Both assertion and reason are correct, and the reason is the correct explanation for the assertion

(B) Both assertion and reason are correct, but the reason is not the correct explanation for the assertion

(C) The assertion is incorrect, but the reason is correct

(D) Both the assertion and reason are incorrect

**Q102.** JEE Main 2015 (Offline)

Which one has the highest boiling point?

(A) Ne (B) Kr (C) Xe (D) He

Open solution

**Q103.** JEE Main 2014 (Online) 19th April Morning Slot

Example of a three-dimensional silicate is :

(A) Zeolites (B) Ultramarines (C) Feldspars (D) All of these

Open solution

**Q104.** JEE Main 2014 (Online) 9th April Morning Slot

The gas evolved on heating $CaF_2$ and $SiO_2$ with concentrated $H_2SO_4$, on hydrolysis gives a white gelatinous precipitate. The precipitate is :

(A) hydrofluosilicic acid (B) silica gel

(C) silicic acid (D) calcium fluorosilicate

Open solution

**Q105.** JEE Main 2014 (Offline)

Which one of the following properties is not shown by NO?

(A) It combines with oxygen to form nitrogen dioxide

(B) It's bond order is 2.5

(C) It is diamagnetic in gaseous state

(D) It is a neutral oxide

Open solution

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**Q106.** JEE Main 2014 (Offline)

The correct statement for the molecule, $CsI_3$ is :

(A) it contains $Cs^{3+}$ and $I^-$ ions.

(B) it contains $Cs^+$, $I^-$ and lattice $I_2$ molecule.

(C) it is a covalent molecule

(D) it contains $Cs^+$ and $I_3^-$ ions

Open solution

**Q107.** JEE Main 2013 (Online) 25th April Morning Slot

The catenation tendency of C, Si and Ge is in the order Ge < Si < C. The bond energies (in kJ $mol^{-1}$) of C-C, Si-Si and Ge-Ge bonds are respectively ;

(A) 348, 297, 260 (B) 297, 348, 260 (C) 348, 260, 297 (D) 260, 297, 348

Open solution

**Q108.** JEE Main 2013 (Online) 23th April Morning Slot

Identify the incorrect statement:

(A) In $(Si_3O_9)^{6-}$, tetrahedral $SiO_4$ units share two oxygen atoms.

(B) Trialkylchlorosilane on hydrolysis gives $R_3SiOH$.

(C) $SiCl_4$ undergoes hydrolysis to give $H_4SiO_4$.

(D) $(Si_3O_9)^{6-}$ has cyclic structure.

Open solution

**Q109.** AIEEE 2012 (Online) 26th May

The number of $S-S$ bonds in $SO_3$, $S_2O_3^{2-}$, $S_2O_6^{2-}$ and $S_2O_8^{2-}$ respectively are

(A) 1, 0, 0, 1 (B) 1, 0, 1, 0 (C) 0, 1, 1, 0 (D) 0, 1, 0, 1

Open solution

**Q110.** AIEEE 2012

The molecule having smallest bond angle is :

(A) $NCl_3$ (B) $AsCl_3$ (C) $SbCl_3$ (D) $PCl_3$

Open solution

**Q111.** AIEEE 2011

Which of the following statements is wrong?

(A) Nitrogen cannot form $d\pi - p\pi$ bond.

(B) Single N – N bond is weaker than the single P – P bond

(C) $N_2O_4$ has two resonance structures

(D) The stability of hydrides increases from $NH_3$ to $BiH_3$ in group 15 of the periodic table

Open solution

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**Q112.** AIEEE 2011

Which of the following statements regarding sulphur is incorrect?

(A) The vapour at 200°C consists mostly of $S_8$ rings

(B) At 600°C the gas mainly consists of $S_2$ molecules

(C) The oxidation state of sulphur is never less than +4 in its compounds

(D) $S_2$ molecule is paramagnetic.

Open solution

**Q113.** AIEEE 2008

Which one of the following is the correct statement?

(A) Boric acid is a protonic acid

(B) Beryllium exhibits coordination number of six

(C) Chlorides of both beryllium and aluminium have bridged chloride structures in solid phase

(D) $B_2H_6.2NH_3$ is known as 'inorganic benzene'

Open solution

**Q114.** AIEEE 2007

Regular use of which of the following fertilizer increases the acidity of soil?

(A) Potassium nitrate

(B) Urea

(C) Superphosphate of lime

(D) Ammonium sulphate

Open solution

**Q115.** AIEEE 2007

The stability of dihalides of $Si, Ge, Sn$ and $Pb$ increases steadily in the sequence :

(A) $PbX_2 << SnX_2 << GeX_2 << SiX_2$

(B) $GeX_2 << SiX_2 << SnX_2 << PbX_2$

(C) $SiX_2 << GeX_2 << PbX_2 << SnX_2$

(D) $SiX_2 << GeX_2 << SnX_2 << PbX_2$

Open solution

**Q116.** AIEEE 2006

Which of the following statements is true?

(A) $H_3PO_3$ is a stronger acid than $H_2SO_3$

(B) In aqueous medium HF is a stronger acid than HCl

(C) $HClO_4$ is a weaker acid than $HClO_3$

(D) $HNO_3$ is a stronger acid than $HNO_2$

Open solution

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**Q117.** AIEEE 2004

Which one the following statement regarding helium is incorrect?

(A) It is used to fill gas balloons instead of hydrogen because it is lighter and non–inflammable

(B) It is used in gas – cooled nuclear reactors

(C) It is used to produce and sustain powerful superconducting reagents

(D) It is used as cryogenic agent for carrying out experiments at low temperatures

Open solution

**Q118.** AIEEE 2004

Beryllium and aluminium exhibit many properties which are similar. But the two elements differ in :

(A) exhibiting maximum covalency in compounds

(B) exhibiting amphoteric nature in their oxides

(C) forming covalent halides

(D) forming polymeric hydrides

Open solution

**Q119.** AIEEE 2003

Glass is a :

(A) super-cooled liquid

(B) gel

(C) polymeric mixture

(D) micro-crystalline solid

Open solution

**Q120.** AIEEE 2003

Which one of the following substances has the highest proton affinity?

(A) $H_2S$

(B) $NH_3$

(C) $PH_3$

(D) $H_2O$

Open solution

**Q121.** AIEEE 2002

Oxidation number of Cl in $CaOCl_2$ (bleaching powder) is :

(A) zero, since it contains $Cl_2$

(B) -1, since it contains $Cl^-$

(C) +1, since it contains $ClO^-$

(D) +1 and -1 since it contains $ClO^-$ and $Cl^-$

Open solution

## Numerical 9 questions

**Q122.** JEE Main 2025 (Online) 28th January Evening Shift

A group 15 element forms $d\pi - d\pi$ bond with transition metals. It also forms hydride, which is a strongest base among the hydrides of other group members that form $d\pi - d\pi$ bond.

The atomic number of the element is _____ .

Open solution

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**Q123.** JEE Main 2024 (Online) 9th April Evening Shift

Number of oxygen atoms present in chemical formula of fuming sulphuric acid is _____.

Open solution

**Q124.** JEE Main 2024 (Online) 1st February Morning Shift

Among the following oxides of p-block elements, number of oxides having amphoteric nature is _____.

$Cl_2O_7, CO, PbO_2, N_2O, NO, Al_2O_3, SiO_2, N_2O_5, SnO_2$

Open solution

**Q125.** JEE Main 2024 (Online) 27th January Evening Shift

1 mole of $PbS$ is oxidised by "X" moles of $O_3$ to get "Y" moles of $O_2$. $X + Y =$ _____.

Open solution

**Q126.** JEE Main 2024 (Online) 27th January Morning Shift

From the given list, the number of compounds with +4 oxidation state of Sulphur _____.

$SO_3, H_2SO_3, SOCl_2, SF_4, BaSO_4, H_2S_2O_7$

Open solution

**Q127.** JEE Main 2023 (Online) 10th April Morning Shift

In the following reactions, the total number of oxygen atoms in X and Y is _____.

$Na_2O + H_2O \rightarrow 2X$

$Cl_2O_7 + H_2O \rightarrow 2Y$

Open solution

**Q128.** JEE Main 2023 (Online) 24th January Evening Shift

Sum of $\pi$-bonds present in peroxodisulphuric acid and pyrosulphuric acid is _____

Open solution

**Q129.** JEE Main 2021 (Online) 24th February Morning Shift

The reaction of sulphur in alkaline medium is given below:

$$S_{8(s)} + a\ OH^-_{(aq)} \longrightarrow b\ S^{2-}_{(aq)} + c\ S_2O_3^{2-}{}_{(aq)} + d\ H_2O_{(l)}$$

The value of 'a' is _____. (Integer answer)

Open solution

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**Q130.** JEE Main 2020 (Online) 6th September Morning Slot

The number of Cl = O bonds in perchloric acid is, "_____".

Open solution

## Out of syllabus 150 questions

*These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.*

**Q131.** JEE Main 2023 (Online) 15th April Morning Shift · MCQ (Single Correct Answer)

For a good quality cement, the ratio of silica to alumina is found to be :

(A) 1.5 (B) 2 (C) 3 (D) 4.5

Open solution

**Q132.** JEE Main 2023 (Online) 13th April Evening Shift · MCQ (Single Correct Answer)

The correct group of halide ions which can be oxidised by oxygen in acidic medium is :

(A) $Br^-$ only

(B) $Cl^-, Br^-$ and $I^-$ only

(C) $Br^-$ and $I^-$ only

(D) $I^-$ only

Open solution

**Q133.** JEE Main 2023 (Online) 13th April Morning Shift · MCQ (Single Correct Answer)

The incorrect statement from the following for borazine is :

(A) It can react with water.

(B) It is a cyclic compound.

(C) It has electronic delocalization.

(D) It contains banana bonds.

Open solution

**Q134.** JEE Main 2023 (Online) 11th April Evening Shift · MCQ (Single Correct Answer)

One mole of $P_4$ reacts with 8 moles of $SOCl_2$ to give 4 moles of A, $x$ mole of $SO_2$ and 2 moles of B. A, B and $x$ respectively are :

(A) $POCl_3$, $S_2Cl_2$ and 4

(B) $PCl_3$, $S_2Cl_2$ and 4

(C) $POCl_3$, $S_2Cl_2$ and 2

(D) $PCl_3$, $S_2Cl_2$ and 2

Open solution

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**Q135.** JEE Main 2023 (Online) 6th April Morning Shift · MCQ (Single Correct Answer)

Open solution

Match List I with List II

| | LIST I<br>Name of reaction | | LIST II<br>Reagent used |
|---|---|---|---|
| A. | Hell-Volhard-Zelinsky reaction | I. | $NaOH + I_2$ |
| B. | Iodoform reaction | II. | (i) $CrO_2Cl_2, CS_2$ (ii) $H_2O$ |
| C. | Etard reaction | III. | (i) $Br_2$ / red phosphorus (ii) $H_2O$ |
| D. | Gatterman-Koch reaction | IV. | $CO, HCl,$, anhyd. $AlCl_3$ |

Choose the correct answer from the options given below:

(A) A-III, B-I, C-IV, D-II

(B) A-III, B-I, C-II, D-IV

(C) A-I, B-II, C-III, D-IV

(D) A-III, B-II, C-I, D-IV

**Q136.** JEE Main 2023 (Online) 1st February Evening Shift · MCQ (Single Correct Answer)

Open solution

The correct order of bond enthalpy ($\mathrm{kJ\ mol^{-1}}$) is :

(A) $\mathrm{C-C > Si-Si > Sn-Sn > Ge-Ge}$

(B) $\mathrm{Si-Si > C-C > Ge-Ge > Sn-Sn}$

(C) $\mathrm{C-C > Si-Si > Ge-Ge > Sn-Sn}$

(D) $\mathrm{Si-Si > C-C > Sn-Sn > Ge-Ge}$

**Q137.** JEE Main 2023 (Online) 1st February Morning Shift · MCQ (Single Correct Answer)

Open solution

Given below are two statements:

Statement I : Chlorine can easily combine with oxygen to form oxides; and the product has a tendency to explode.

Statement II : Chemical reactivity of an element can be determined by its reaction with oxygen and halogens.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is true but Statement II is false

(B) Both the Statements I and II are false

(C) Statement I is false but Statement II is true

(D) Both the Statements I and II are true

**Q138.** JEE Main 2023 (Online) 31st January Evening Shift · MCQ (Single Correct Answer)

Open solution

The Lewis acid character of boron tri halides follows the order :

(A) $\mathrm{BI_3 > BBr_3 > BCl_3 > BF_3}$

(B) $\mathrm{BCl_3 > BF_3 > BBr_3 > BI_3}$

(C) $\mathrm{BF_3 > BCl_3 > BBr_3 > BI_3}$

(D) $\mathrm{BBr_3 > BI_3 > BCl_3 > BF_3}$

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**Q139.** JEE Main 2023 (Online) 30th January Evening Shift · MCQ (Single Correct Answer)

Boric acid is solid, whereas $BF_3$ is gas at room temperature because of :

(A) Strong van der Waal's interaction in Boric acid

(B) Strong covalent bond in $BF_3$

(C) Strong hydrogen bond in Boric acid

(D) Strong ionic bond in Boric acid

Open solution

**Q140.** JEE Main 2023 (Online) 29th January Morning Shift · MCQ (Single Correct Answer)

During the borax bead test with $CuSO_4$, a blue green colour of the bead was observed in oxidising flame due to the formation of :

(A) CuO (B) $Cu(BO_2)_2$ (C) $Cu_3B_2$ (D) Cu

Open solution

**Q141.** JEE Main 2023 (Online) 29th January Morning Shift · MCQ (Single Correct Answer)

"A" obtained by Ostwald's method involving air oxidation of $NH_3$, upon further air oxidation produces "B", "B" on hydration forms an oxoacid of Nitrogen along with evolution of "A". The oxoacid also produces "A" and gives positive brown ring test.

Identify A and B, respectively :

(A) $N_2O_3, NO_2$ (B) $NO_2, N_2O_5$ (C) $NO_2, N_2O_4$ (D) $NO, NO_2$

Open solution

**Q142.** JEE Main 2023 (Online) 25th January Evening Shift · MCQ (Single Correct Answer)

Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason R

Assertion A : Carbon forms two important oxides - CO and $CO_2$. CO is neutral whereas $CO_2$ is acidic in nature

Reason R : $CO_2$ can combine with water in a limited way to form carbonic acid, while CO is sparingly soluble in water

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) A is not correct but R is correct

(B) Both A and R are correct but R is NOT the correct explanation of A

(C) Both A and R are correct and R is the correct explanation of A

(D) A is correct but R is not correct

Open solution

**Q143.** JEE Main 2023 (Online) 25th January Morning Shift · MCQ (Single Correct Answer)

Reaction of thionyl chloride with white phosphorus forms a compound [A], which on hydrolysis gives [B], a dibasic acid. [A] and [B] are respectively :

(A) $P_4O_6$ and $H_3PO_3$ (B) $POCl_3$ and $H_3PO_4$ (C) $PCl_5$ and $H_3PO_4$ (D) $PCl_3$ and $H_3PO_3$

Open solution

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**Q144.** JEE Main 2023 (Online) 25th January Morning Shift · MCQ (Single Correct Answer)

Compound A reacts with $NH_4Cl$ and forms a compound B. Compound B reacts with $H_2O$ and excess of $CO_2$ to form compound C which on passing through or reaction with saturated NaCl solution forms sodium hydrogen carbonate. Compound A, B and C, are respectively :

(A) $Ca(OH)_2, NH_4^{\oplus}, (NH_4)_2CO_3$
(B) $CaCl_2, NH_4^{\oplus}, (NH_4)_2CO_3$
(C) $CaCl_2, NH_3, NH_4HCO_3$
(D) $Ca(OH)_2, NH_3, NH_4HCO_3$

Open solution

**Q145.** JEE Main 2023 (Online) 24th January Morning Shift · MCQ (Single Correct Answer)

Which of the Phosphorus oxoacid can create silver mirror from $AgNO_3$ solution?

(A) $H_4P_2O_5$
(B) $H_4P_2O_7$
(C) $(HPO_3)_n$
(D) $H_4P_2O_6$

Open solution

**Q146.** JEE Main 2022 (Online) 29th July Evening Shift · MCQ (Single Correct Answer)

When borax is heated with $CoO$ on a platinum loop, blue coloured bead formed is largely due to :

(A) $B_2O_3$
(B) $Co(BO_2)_2$
(C) $CoB_4O_7$
(D) $Co\,[B_4O_5(OH)_4]$

Open solution

**Q147.** JEE Main 2022 (Online) 29th July Evening Shift · MCQ (Single Correct Answer)

Dinitrogen is a robust compound, but reacts at high altitudes to form oxides. The oxide of nitrogen that can damage plant leaves and retard photosynthesis is :

(A) $NO$
(B) $NO_3^-$
(C) $NO_2$
(D) $NO_2^-$

Open solution

**Q148.** JEE Main 2022 (Online) 28th July Evening Shift · MCQ (Single Correct Answer)

White phosphorus reacts with thionyl chloride to give :

(A) $PCl_5, SO_2$ and $S_2Cl_2$
(B) $PCl_3, SO_2$ and $S_2Cl_2$
(C) $PCl_3, SO_2$ and $Cl_2$
(D) $PCl_5, SO_2$ and $Cl_2$

Open solution

**Q149.** JEE Main 2022 (Online) 28th July Evening Shift · MCQ (Single Correct Answer)

Concentrated $HNO_3$ reacts with Iodine to give :

(A) HI, $NO_2$ and $H_2O$
(B) $HIO_2$, $N_2O$ and $H_2O$
(C) $HIO_3, NO_2$ and $H_2O$
(D) $HIO_4, NO_2$ and $H_2O$

Open solution

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**Q150.** JEE Main 2022 (Online) 27th July Evening Shift · MCQ (Single Correct Answer)

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Boron is unable to form $BF_6^{3-}$.

Reason (R) : Size of B is very small.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both (A) and (R) are true and (R) is the correct explanation of (A)

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(C) (A) is true but (R) is false

(D) (A) is false but (R) is true

**Q151.** JEE Main 2022 (Online) 27th July Morning Shift · MCQ (Single Correct Answer)

Open solution

Given below are two statements.

Statement I: The chlorides of $Be$ and $Al$ have Cl-bridged structure. Both are soluble in organic solvents and act as Lewis bases.

Statement II: Hydroxides of $Be$ and $Al$ dissolve in excess alkali to give beryllate and aluminate ions.

In the light of the above statements, choose the correct answer from the options given below.

(A) Both Statement I and Statement II are true.

(B) Both Statement I and Statement II are false.

(C) Statement I is true but Statement II is false.

(D) Statement I is false but Statement II is true.

**Q152.** JEE Main 2022 (Online) 27th July Morning Shift · MCQ (Single Correct Answer)

Open solution

Which oxoacid of phosphorous has the highest number of oxygen atoms present in its chemical formula?

(A) Pyrophosphorus acid

(B) Hypophosphoric acid

(C) Phosphoric acid

(D) Pyrophosphoric acid

**Q153.** JEE Main 2022 (Online) 26th July Evening Shift · MCQ (Single Correct Answer)

Open solution

The metal that has very low melting point and its periodic position is closer to a metalloid is :

(A) Al

(B) Ga

(C) Se

(D) In

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**Q154.** JEE Main 2022 (Online) 26th July Evening Shift · MCQ (Single Correct Answer)

Open solution

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion : Boric acid is a weak acid.

Reason R : Boric acid is not able to release $H^+$ ion on its own. It receives $OH^-$ ion from water and releases $H^+$ ion.

In the light of the above statements, choose the most appropriate answer from the options given below.

**(A)** Both A and R are true and R is the correct explanation of A.

**(B)** Both A and R are true but R is NOT the correct explanation of A.

**(C)** A is correct but R is not correct.

**(D)** A is not correct but R is correct.

**Q155.** JEE Main 2022 (Online) 26th July Morning Shift · MCQ (Single Correct Answer)

Open solution

Borazine, also known as inorganic benzene, can be prepared by the reaction of 3-equivalents of "$X$" with 6-equivalents of "$Y$". "$X$" and "$Y$", respectively are :

**(A)** $B(OH)_3$ and $NH_3$

**(B)** $B_2H_6$ and $NH_3$

**(C)** $B_2H_6$ and $HN_3$

**(D)** $NH_3$ and $B_2O_3$

**Q156.** JEE Main 2022 (Online) 25th July Evening Shift · MCQ (Single Correct Answer)

Open solution

The total number of acidic oxides from the following list is

NO, $N_2O$, $B_2O_3$, $N_2O_5$, CO, $SO_3$, $P_4O_{10}$

**(A)** 3 **(B)** 4 **(C)** 5 **(D)** 6

**Q157.** JEE Main 2022 (Online) 25th July Morning Shift · MCQ (Single Correct Answer)

Open solution

Match List I with List II

| | List I | | List II |
|---|---|---|---|
| (A) | $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$ | (I) | Cu |
| (B) | $CO(g) + 3H_2(g) \rightarrow CH_4(g) + H_2O(g)$ | (II) | $Cu/ZnO - Cr_2O_3$ |
| (C) | $CO(g) + H_2(g) \rightarrow HCHO(g)$ | (III) | $Fe_xO_y + K_2O + Al_2O_3$ |
| (D) | $CO(g) + 2H_2(g) \rightarrow CH_3OH(g)$ | (IV) | Ni |

Choose the correct answer from the options given below :

**(A)** $(A) - (II), (B) - (IV), (C) - (I), (D) - (III)$

**(B)** $(A) - (II), (B) - (I), (C) - (IV), (D) - (III)$

**(C)** $(A) - (III), (B) - (IV), (C) - (I), (D) - (II)$

**(D)** $(A) - (III), (B) - (I), (C) - (IV), (D) - (II)$

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**Q158.** JEE Main 2022 (Online) 25th July Morning Shift · MCQ (Single Correct Answer)

The geometry around boron in the product 'B' formed from the following reaction is

$$BF_3 + NaH \xrightarrow{450\ K} A + NaF$$

$$A + NMe_3 \rightarrow B$$

(A) trigonal planar (B) tetrahedral (C) pyramidal (D) square planar

Open solution

**Q159.** JEE Main 2022 (Online) 25th July Morning Shift · MCQ (Single Correct Answer)

The interhalogen compound formed from the reaction of bromine with excess of fluorine is a :

(A) hypohalite (B) halate (C) perhalate (D) halite

Open solution

**Q160.** JEE Main 2022 (Online) 29th June Evening Shift · MCQ (Single Correct Answer)

Aqueous solution of which of the following boron compounds will be strongly basic in nature?

(A) $NaBH_4$ (B) $LiBH_4$ (C) $B_2H_6$ (D) $Na_2B_4O_7$

Open solution

**Q161.** JEE Main 2022 (Online) 29th June Morning Shift · MCQ (Single Correct Answer)

Match List-I with List-II.

| | List-I (Metal) | | List-II (Application) |
|---|---|---|---|
| (A) | Cs | (I) | High temperature thermometer |
| (B) | Ga | (II) | Water repellent sprays |
| (C) | B | (III) | Photoelectric cells |
| (D) | Si | (IV) | Bullet proof vest |

Choose the most appropriate answer from the options given below :

(A) (A)-(III), (B)-(I), (C)-(IV), (D)-(II) (B) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

(C) (A)-(II), (B)-(III), (C)-(IV), (D)-(I) (D) (A)-(I), (B)-(IV), (C)-(II), (D)-(III)

Open solution

**Q162.** JEE Main 2022 (Online) 29th June Morning Shift · MCQ (Single Correct Answer)

The oxoacid of phosphorus that is easily obtained from a reaction of alkali and white phosphorus and has two P-H bonds, is :

(A) Phosphonic acid (B) Phosphinic acid

(C) Pyrophosphorus acid (D) Hypophosphoric acid

Open solution

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**Q163.** JEE Main 2022 (Online) 28th June Evening Shift · MCQ (Single Correct Answer)

Among the given oxides of nitrogen ; $N_2O$, $N_2O_3$, $N_2O_4$ and $N_2O_5$, the number of compound/(s) having N - N bond is :

(A) 1 (B) 2 (C) 3 (D) 4

Open solution

**Q164.** JEE Main 2022 (Online) 28th June Evening Shift · MCQ (Single Correct Answer)

Which of the following oxoacids of sulphur contains "S" in two different oxidation states?

(A) $H_2S_2O_3$ (B) $H_2S_2O_6$ (C) $H_2S_2O_7$ (D) $H_2S_2O_8$

Open solution

**Q165.** JEE Main 2022 (Online) 28th June Morning Shift · MCQ (Single Correct Answer)

Given below are two statements :

Statement I : The pentavalent oxide of group-15 element, $E_2O_5$, is less acidic than trivalent oxide, $E_2O_3$, of the same element.

Statement II : The acidic character of trivalent oxide of group 15 elements, $E_2O_3$, decreases down the group.

In light of the above statements, choose most appropriate answer from the options given below :

(A) Both Statement I and Statement II are true.

(B) Both Statement I and Statement II are false.

(C) Statement I is true, but Statement II is false.

(D) Statement I is false, but Statement II is true.

Open solution

**Q166.** JEE Main 2022 (Online) 27th June Evening Shift · MCQ (Single Correct Answer)

The gas produced by treating an aqueous solution of ammonium chloride with sodium nitrite is :

(A) $NH_3$ (B) $N_2$ (C) $N_2O$ (D) $Cl_2$

Open solution

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**Q167.** JEE Main 2022 (Online) 27th June Morning Shift · MCQ (Single Correct Answer)

Open solution

Match List - I with List - II.

| | List - I (Si-Compounds) | | List - II (Si-Polymeric/Other Products |
|---|---|---|---|
| (A) | $(CH_3)_4Si$ | (I) | Chain Silicone |
| (B) | $(CH_3)Si(OH)_3$ | (II) | Dimeric Silicone |
| (C) | $(CH_3)_2Si(OH)_2$ | (III) | Silane |
| (D) | $(CH_3)_3Si(OH)$ | (IV) | 2D - Silicone |

Choose the correct answer from the options given below :

**(A)** (A) - (III), (B) - (II), (C) - (I), (D) - (IV)

**(B)** (A) - (IV), (B) - (I), (C) - (II), (D) - (III)

**(C)** (A) - (II), (B) - (I), (C) - (IV), (D) - (III)

**(D)** (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

**Q168.** JEE Main 2022 (Online) 27th June Morning Shift · MCQ (Single Correct Answer)

Open solution

Heating white phosphorus with conc. NaOH solution gives mainly :

**(A)** $Na_3P$ and $H_2O$

**(B)** $H_3PO$ and NaH

**(C)** $P(OH)_3$ and $NaH_2PO_4$

**(D)** $PH_3$ and $NaH_2PO_2$

**Q169.** JEE Main 2022 (Online) 26th June Morning Shift · MCQ (Single Correct Answer)

Open solution

The correct order of melting points of hydrides of group 16 elements is :

**(A)** $H_2S < H_2Se < H_2Te < H_2O$

**(B)** $H_2O < H_2S < H_2Se < H_2Te$

**(C)** $H_2S < H_2Te < H_2Se < H_2O$

**(D)** $H_2Se < H_2S < H_2Te < H_2O$

**Q170.** JEE Main 2022 (Online) 26th June Morning Shift · MCQ (Single Correct Answer)

Open solution

Consider the following reaction :

A + alkali $\rightarrow$ B (Major Product)

If B is an oxoacid of phosphorus with no P-H bond, then A is :

**(A)** White $P_4$

**(B)** Red $P_4$

**(C)** $P_2O_3$

**(D)** $H_3PO_3$

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**Q171.** JEE Main 2022 (Online) 25th June Evening Shift · MCQ (Single Correct Answer)

The number of bridged oxygen atoms present in compound B formed from the following reactions is

$Pb(NO_3)_2 \xrightarrow{673\ K} A + PbO + O_2$

$A \xrightarrow{Dimerise} B$

(A) 0 (B) 1 (C) 2 (D) 3

Open solution

**Q172.** JEE Main 2022 (Online) 24th June Morning Shift · MCQ (Single Correct Answer)

Identify the correct statement for $B_2H_6$ from those given below :

(A) In $B_2H_6$, all B-H bonds are equivalent.

(B) In $B_2H_6$, there are four 3-centre-2-electron bonds.

(C) $B_2H_6$ is a Lewis acid.

(D) $B_2H_6$ can be synthesized from both $BF_3$ and $NaBH_4$.

(E) $B_2H_6$ is a planar molecule.

Choose the most appropriate answer from the options given below :

(A) (A) and (E) only (B) (B), (C) and (E) only

(C) (C) and (D) only (D) (C) and (E) only

Open solution

**Q173.** JEE Main 2022 (Online) 24th June Morning Shift · MCQ (Single Correct Answer)

Which one of the following elemental forms is not present in the enamel of the teeth?

(A) $Ca^{2+}$ (B) $P^{3+}$ (C) $F^-$ (D) $P^{5+}$

Open solution

**Q174.** JEE Main 2021 (Online) 1st September Evening Shift · MCQ (Single Correct Answer)

The oxide without nitrogen-nitrogen bond is :

(A) $N_2O$ (B) $N_2O_4$ (C) $N_2O_3$ (D) $N_2O_5$

Open solution

**Q175.** JEE Main 2021 (Online) 31st August Evening Shift · MCQ (Single Correct Answer)

Which one of the following correctly represents the order of stability of oxides, $X_2O$; (X = halogen)?

(A) Br > Cl > I (B) Br > I > Cl (C) Cl > I > Br (D) I > Cl > Br

Open solution

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**Q176.** JEE Main 2021 (Online) 31st August Evening Shift · MCQ (Single Correct Answer)

The number of S = O bonds present in sulphurous acid, peroxodisulphuric acid and pyrosulphuric acid, respectively are :

(A) 2, 3 and 4 (B) 1, 4 and 3 (C) 2, 4 and 3 (D) 1, 4 and 4

Open solution

**Q177.** JEE Main 2021 (Online) 27th August Evening Shift · MCQ (Single Correct Answer)

Which one of the following is formed (mainly) when red phosphorus is heated in a sealed tube at 803 K?

(A) White phosphorus (B) Yellow phosphorus

(C) $\beta$-Black phosphorus (D) $\alpha$-Black phosphorus

Open solution

**Q178.** JEE Main 2021 (Online) 27th August Morning Shift · MCQ (Single Correct Answer)

In which one of the following molecules strongest back donation of an electron pair from halide to boron is expected?

(A) $BCl_3$ (B) $BF_3$ (C) $BBr_3$ (D) $BI_3$

Open solution

**Q179.** JEE Main 2021 (Online) 26th August Evening Shift · MCQ (Single Correct Answer)

The number of non-ionisable hydrogen atoms present in the final product obtained from the hydrolysis of $PCl_5$ is :

(A) 0 (B) 2 (C) 1 (D) 3

Open solution

**Q180.** JEE Main 2021 (Online) 26th August Evening Shift · MCQ (Single Correct Answer)

Chalcogen group elements are :

(A) Se, Tb and Pu. (B) Se, Te and Po. (C) S, Te and Pm. (D) O, Ti and Po.

Open solution

**Q181.** JEE Main 2021 (Online) 26th August Morning Shift · MCQ (Single Correct Answer)

The incorrect statement is :

(A) $Cl_2$ is more reactive than ClF.

(B) $F_2$ is more reactive than ClF.

(C) On hydrolysis ClF forms HOCl and HF.

(D) $F_2$ is a stronger oxidizing agent than $Cl_2$ in aqueous solution.

Open solution

**Q182.** JEE Main 2021 (Online) 27th July Evening Shift · MCQ (Single Correct Answer)

Number of Cl = O bonds in chlorous acid, chloric acid and perchloric acid respectively are :

(A) 3, 1 and 1 (B) 4, 1 and 0 (C) 1, 1 and 3 (D) 1, 2 and 3

Open solution

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**Q183.** JEE Main 2021 (Online) 27th July Morning Shift · MCQ (Single Correct Answer)

The product obtained from the electrolytic oxidation of acidified sulphate solutions, is :

(A) $HSO_4^-$ (B) $HO_3SOOSO_3H$ (C) $HO_2SOSO_2H$ (D) $HO_3SOSO_3H$

Open solution

**Q184.** JEE Main 2021 (Online) 22th July Evening Shift · MCQ (Single Correct Answer)

Given below are the statements about diborane

(a) Diborane is prepared by the oxidation of $NaBH_4$ with $I_2$

(b) Each boron atom is in $sp^2$ hybridized state

(c) Diborane has one bridged 3 centre-2-electron bond

(d) Diborane is a planar molecule

The option with correct statements is :

(A) (c) and (d) only (B) (a) only (C) (c) only (D) (a) and (b) only

Open solution

**Q185.** JEE Main 2021 (Online) 20th July Morning Shift · MCQ (Single Correct Answer)

Chemical nature of the nitrogen oxide compound obtained from a reaction of concentrated nitric acid and $P_4O_{10}$ (in 4 : 1 ratio) is :

(A) acidic (B) basic (C) amphoteric (D) neutral

Open solution

**Q186.** JEE Main 2021 (Online) 18th March Morning Shift · MCQ (Single Correct Answer)

The number of ionisable hydrogens present in the product obtained from a reaction of phosphorus trichloride and phosphonic acid is :

(A) 3 (B) 0 (C) 2 (D) 1

Open solution

**Q187.** JEE Main 2021 (Online) 17th March Evening Shift · MCQ (Single Correct Answer)

The set that represents the pair of neutral oxides of nitrogen is :

(A) NO and $N_2O$ (B) $N_2O$ and $NO_2$ (C) NO and $NO_2$ (D) $N_2O$ and $N_2O_3$

Open solution

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**Q188.** JEE Main 2021 (Online) 16th March Evening Shift · MCQ (Single Correct Answer)

Open solution

The INCORRECT statement regarding the structure of $C_{60}$ is :

(A) The five-membered rings are fused only to six-membered rings.

(B) It contains 12 six-membered rings and 24 five-membered rings.

(C) Each carbon atom forms three sigma bonds.

(D) The six-membered rings are fused to both six and five-membered rings.

**Q189.** JEE Main 2021 (Online) 16th March Morning Shift · MCQ (Single Correct Answer)

Open solution

Match List - I with List - II :

| | **List - I Industrial process** | | **List - II Application** |
|---|---|---|---|
| (a) | Haber's process | (i) | $HNO_3$ synthesis |
| (b) | Ostwald's process | (ii) | Aluminium extraction |
| (c) | Contact process | (iii) | $NH_3$ synthesis |
| (d) | Hal-Heroult process | (iv) | $H_2SO_4$ synthesis |

Choose the correct answer from the options given below :

(A) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

(B) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

(C) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

(D) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

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**Q190.** JEE Main 2021 (Online) 16th March Morning Shift · MCQ (Single Correct Answer)

Open solution

Match List - I with List - II :

| | List - I<br>Name of oxo acid | | List - II<br>Oxidation state of 'P' |
|---|---|---|---|
| (a) | Hypophosphorous acid | (i) | +5 |
| (b) | Orthophosphoric acid | (ii) | +4 |
| (c) | Hypophosphoric acid | (iii) | +3 |
| (d) | Orthophosphorous acid | (iv) | +2 |
| | | (v) | +1 |

Choose the correct answer from the options given below :

**(A)** (a) - (iv), (b) - (v), (c) - (ii), (d) - (iii)

**(B)** (a) - (iv), (b) - (i), (c) - (ii), (d) - (iii)

**(C)** (a) - (v), (b) - (i), (c) - (ii), (d) - (iii)

**(D)** (a) - (v), (b) - (iv), (c) - (ii), (d) - (iii)

**Q191.** JEE Main 2021 (Online) 26th February Morning Shift · MCQ (Single Correct Answer)

Open solution

On treating a compound with warm dil. $H_2SO_4$, gas X is evolved which turns $K_2Cr_2O_7$ paper acidified with dil. $H_2SO_4$ to a green compound Y. X and Y respectively are :

**(A)** X = $SO_2$, Y = $Cr_2O_3$

**(B)** X = $SO_3$, Y = $Cr_2(SO_4)_3$

**(C)** X = $SO_3$, Y = $Cr_2O_3$

**(D)** X = $SO_2$, Y = $Cr_2(SO_4)_3$

**Q192.** JEE Main 2021 (Online) 25th February Evening Shift · MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : $\alpha$ and $\beta$ forms of sulphur can change reversibly between themselves with slow heating or slow cooling.

Statement II : At room temperature the stable crystalline form of sulphur is monoclinic sulphur.

In the light of the above statements, choose the correct answer from the options given below :

**(A)** Statement I is true but Statement II is false.

**(B)** Statement I is false but Statement II is true.

**(C)** Both Statement I and Statement II are true.

**(D)** Both Statement I and Statement II are false.

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**Q193.** JEE Main 2021 (Online) 25th February Morning Shift · MCQ (Single Correct Answer)

The correct statement about $B_2H_6$ is :

(A) Its fragment, $BH_3$, behaves as a Lewis base.

(B) All B–H–B angles are of 120°.

(C) The two B–H–B bonds are not of same length.

(D) Terminal B–H bonds have less p-character when compared to bridging bonds.

Open solution

**Q194.** JEE Main 2020 (Online) 6th September Evening Slot · MCQ (Single Correct Answer)

The reaction of NO with $N_2O_4$ at 250 K gives :

(A) $N_2O_3$ (B) $N_2O_5$ (C) $N_2O$ (D) $NO_2$

Open solution

**Q195.** JEE Main 2020 (Online) 6th September Evening Slot · MCQ (Single Correct Answer)

Reaction of an inorganic sulphite X with dilute $H_2SO_4$ generates compound Y. Reaction of Y with NaOH gives X. Further, the reaction of X with Y and water affords compound Z. Y and Z, respectively, are :

(A) S and $Na_2SO_3$ (B) $SO_2$ and $NaHSO_3$ (C) $SO_2$ and $Na_2SO_3$ (D) $SO_3$ and $NaHSO_3$

Open solution

**Q196.** JEE Main 2020 (Online) 5th September Evening Slot · MCQ (Single Correct Answer)

Reaction of ammonia with excess $Cl_2$ gives :

(A) $NH_4Cl$ and $N_2$ (B) $NH_4Cl$ and HCl (C) $NCl_3$ and HCl (D) $NCl_3$ and $NH_4Cl$

Open solution

**Q197.** JEE Main 2020 (Online) 4th September Morning Slot · MCQ (Single Correct Answer)

On heating, lead (II) nitrate gives a brown gas (A). The gas (A) on cooling changes to a colourless solid/liquid (B). (B) on heating with NO changes to a blue solid (C). The oxidation number of nitrogen in solid (C) is :

(A) +3 (B) +4 (C) +5 (D) +2

Open solution

**Q198.** JEE Main 2020 (Online) 3rd September Morning Slot · MCQ (Single Correct Answer)

Aqua regia is used for dissolving noble metals (Au, Pt, etc.). The gas evolved in this process is :

(A) $N_2O_3$ (B) $N_2$ (C) $N_2O_5$ (D) NO

Open solution

**Q199.** JEE Main 2020 (Online) 3rd September Morning Slot · MCQ (Single Correct Answer)

In a molecule of pyrophosphoric acid, the number of P - OH, P = O and P - O - P bonds/moiety(ies) respectively are :

(A) 4, 2 and 0 (B) 4, 2 and 1 (C) 3, 3 and 3 (D) 2, 4 and 1

Open solution

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**Q200.** JEE Main 2020 (Online) 2nd September Morning Slot · MCQ (Single Correct Answer)

On heating compound (A) gives a gas (B) which is a constituent of air. This gas when treated with $H_2$ in the presence of a catalyst gives another gas (C) which is basic in nature. (A) should **not** be :

(A) $Pb(NO_3)_2$ (B) $(NH_4)_2Cr_2O_7$ (C) $NH_4NO_2$ (D) $NaN_3$

Open solution

**Q201.** JEE Main 2020 (Online) 9th January Morning Slot · MCQ (Single Correct Answer)

'X' melts at low temperature and is a bad conductor of electricity in both liquid and solid state. X is :

(A) Mercury (B) Carbon tetrachloride

(C) Zinc sulphide (D) Silicon carbide

Open solution

**Q202.** JEE Main 2020 (Online) 8th January Evening Slot · MCQ (Single Correct Answer)

White Phosphorus on reaction with concentrated NaOH solution in an inert atmosphere of $CO_2$ gives phosphine and compound (X). (X) on acidification with HCl gives compound (Y). The basicity of compound (Y) is :

(A) 2 (B) 4 (C) 1 (D) 3

Open solution

**Q203.** JEE Main 2020 (Online) 8th January Morning Slot · MCQ (Single Correct Answer)

The number of bonds between sulphur and oxygen atoms in $S_2O_8^{2-}$ and the number of bonds between sulphur and sulphur atoms in rhombic sulphur, respectively, are :

(A) 4 and 8 (B) 8 and 6 (C) 4 and 6 (D) 8 and 8

Open solution

**Q204.** JEE Main 2020 (Online) 7th January Evening Slot · MCQ (Single Correct Answer)

In the following reaction, products (A) and (B) respectively, are :

$NaOH + Cl_2 \rightarrow$ (A) + side products
(hot and conc.)

$Ca(OH)_2 + Cl_2 \rightarrow$ (B) + side products (dry)

(A) NaOCl and $Ca(OCl)_2$ (B) $NaClO_3$ and $Ca(ClO_3)_2$

(C) NaOCl and $Ca(ClO_3)_2$ (D) $NaClO_3$ and $Ca(OCl)_2$

Open solution

**Q205.** JEE Main 2019 (Online) 12th April Evening Slot · MCQ (Single Correct Answer)

The C–C bond length is maximum in :

(A) $C_{60}$ (B) diamond (C) graphite (D) $C_{70}$

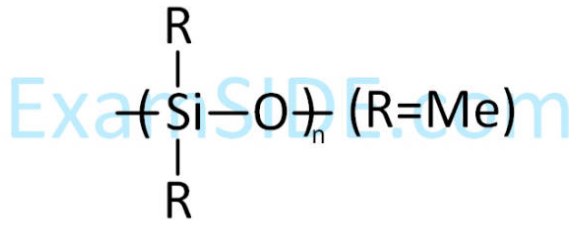
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**Q206.** JEE Main 2019 (Online) 12th April Morning Slot · MCQ (Single Correct Answer)

The basic structural unit of feldspar, zeolites, mica, and asbestos is :

(A)



(B) $(SiO_4)^{4-}$

(C) $(SiO_3)^{2-}$

(D) $SiO_2$

Open solution

**Q207.** JEE Main 2019 (Online) 10th April Evening Slot · MCQ (Single Correct Answer)

The number of pentagons in $C_{60}$ and trigons (triangles) in white phosphorus, respectively, are :

(A) 12 and 3 (B) 20 and 3 (C) 20 and 4 (D) 12 and 4

Open solution

**Q208.** JEE Main 2019 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

The correct order of catenation is :

(A) C > Sn > Si ≈ Ge (B) Si > Sn > C > Ge

(C) C > Si > Ge ≈ Sn (D) Ge > Sn > Si > C

Open solution

**Q209.** JEE Main 2019 (Online) 9th April Evening Slot · MCQ (Single Correct Answer)

The amorphous form of silica is :

(A) kieselguhr (B) quartz (C) tridymite (D) cristobalite

Open solution

**Q210.** JEE Main 2019 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

$C_{60}$ an allotrope of carbon contains :

(A) 12 hexagons and 20 pentagons. (B) 20 hexagons and 12 pentagons.

(C) 16 hexagons and 16 pentagons (D) 18 hexagons and 14 pentagons.

Open solution

**Q211.** JEE Main 2019 (Online) 8th April Morning Slot · MCQ (Single Correct Answer)

Diborane ($B_2H_6$) reacts with $O_2$ and $H_2O$ independently, then products formed are :

(A) $B_2O_3$ and $HBO_2$ (B) $B_2O_3$ and $H_3BO_3$ (C) $HBO_2$ and $H_3BO_3$ (D) $HBO_2$ and $HBO_3$

Open solution

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**Q212.** JEE Main 2019 (Online) 12th January Evening Slot · MCQ (Single Correct Answer)

Chlorine on reaction with hot and concentrated sodium hydroxide give :

(A) $C\ell O_3^-$ and $C\ell O_2^-$

(B) $C\ell^-$ and $C\ell O^-$

(C) $C\ell^-$ and $C\ell O_2^-$

(D) $C\ell^-$ and $C\ell O_3^-$

Open solution

**Q213.** JEE Main 2019 (Online) 12th January Morning Slot · MCQ (Single Correct Answer)

Iodine reacts with concentrated $HNO_3$ to yield Y along with other products. The oxidation state of iodine in Y, is :

(A) 3 (B) 1 (C) 7 (D) 5

Open solution

**Q214.** JEE Main 2019 (Online) 11th January Evening Slot · MCQ (Single Correct Answer)

The hydride that is NOT electron deficient :

(A) $B_2H_6$ (B) $AlH_3$ (C) $GaH_3$ (D) $SiH_4$

Open solution

**Q215.** JEE Main 2019 (Online) 11th January Evening Slot · MCQ (Single Correct Answer)

The relative stability of +1 oxidation state of group 13 elements follows the order :

(A) Tl < In < Ga < Al

(B) Al < Ga < Tl < In

(C) Al < Ga < In < Tl

(D) Ga < Al < In < Tl

Open solution

**Q216.** JEE Main 2019 (Online) 11th January Morning Slot · MCQ (Single Correct Answer)

The chloride that CANNOT get hydrolysed is :

(A) $SnCl_4$ (B) $SiCl_4$ (C) $PbCl_4$ (D) $CCl_4$

Open solution

**Q217.** JEE Main 2019 (Online) 10th January Evening Slot · MCQ (Single Correct Answer)

Among the following reactions of hydrogen with halogens, the one that requires a catalyst is :

(A) $H_2 + Br_2 \rightarrow 2HBr$

(B) $H_2 + Cl_2 \rightarrow 2HCl$

(C) $H_2 + F_2 \rightarrow 2HF$

(D) $H_2 + I_2 \rightarrow 2HI$

Open solution

**Q218.** JEE Main 2019 (Online) 10th January Evening Slot · MCQ (Single Correct Answer)

The number of 2-centre-2-electron and 3-centre -2-electron bonds in $B_2H_6$, respectively, are :

(A) 2 and 2 (B) 2 and 1 (C) 2 and 4 (D) 4 and 2

Open solution

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**Q219.** JEE Main 2019 (Online) 9th January Evening Slot · MCQ (Single Correct Answer)

Good reducing nature of $H_3PO_2$ is attributed to the presence of :

(A) Two P – OH bonds
(B) One P – H bond
(C) Two P – H bonds
(D) One P – OH bond

Open solution

**Q220.** JEE Main 2019 (Online) 9th January Morning Slot · MCQ (Single Correct Answer)

Correct statements among a to d regarding silicones are :

(a) They are polymers with hydrophobic character.
(b) They are biocompatible.
(c) In general, they have high thermal stability and low dielectric strength.
(d) Usually, they are resistant to oxidation and used as greases.

(A) (a), (b), (c) and (d)
(B) (a), (b) and (c) only
(C) (a) and (b) only
(D) (a), (b) and (d) only

Open solution

**Q221.** JEE Main 2018 (Online) 16th April Morning Slot · MCQ (Single Correct Answer)

Among the oxides of nitrogen : $N_2O_3$, $N_2O_4$ and $N_2O_5$; the molecule(s) having nitrogen-nitrogen bond is / are :

(A) Only $N_2O_5$
(B) $N_2O_3$ and $N_2O_5$
(C) $N_2O_4$ and $N_2O_5$
(D) $N_2O_3$ and $N_2O_4$

Open solution

**Q222.** JEE Main 2018 (Online) 16th April Morning Slot · MCQ (Single Correct Answer)

A group 13 element 'X' reacts with chlorine gas to produce a compound $XCl_3$. $XCl_3$ is electron deficient and easily reacts with $NH_3$ to form $Cl_3X \leftarrow NH_3$ adduct ; however, $XCl_3$ does not dimerize X is :

(A) B
(B) Al
(C) Ga
(D) In

Open solution

**Q223.** JEE Main 2018 (Offline) · MCQ (Single Correct Answer)

When metal 'M' is treated with NaOH, a white gelatinous precipitate 'X' is obtained, which is soluble in excess of NaOH. Compound 'X' when heated strongly gives an oxide which is used in chromatography as an adsorbent. The metal 'M' is

(A) Fe
(B) Zn
(C) Ca
(D) Al

Open solution

**Q224.** JEE Main 2018 (Online) 15th April Evening Slot · MCQ (Single Correct Answer)

The number of P – O bonds in $P_4O_6$ is :

(A) 6
(B) 9
(C) 12
(D) 18

Open solution

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**Q225.** JEE Main 2018 (Online) 15th April Evening Slot · MCQ (Single Correct Answer)

Lithium aluminium hydride reacts with silicon tetrachloride to form :

(A) LiCl, $AlH_3$ and $SiH_4$

(B) LiCl, $AlCl_3$ and $SiH_4$

(C) LiH, $AlCl_3$ and $SiCl_2$

(D) LiH, $AlH_3$ and $SiH_4$

Open solution

**Q226.** JEE Main 2018 (Online) 15th April Evening Slot · MCQ (Single Correct Answer)

In $XeO_3F_2$, the number of bond pair(s), $\pi$-bond(s) and lone pair(s) on Xe atom respectively are :

(A) 5, 2, 0

(B) 4, 2, 2

(C) 5, 3, 0

(D) 4, 4, 0

Open solution

**Q227.** JEE Main 2018 (Online) 15th April Morning Slot · MCQ (Single Correct Answer)

Xenon hexafluoride on partial hydrolysis produces compounds 'X' and 'Y' . Compounds 'X' and 'Y' and the oxidation state of Xe are respectively :

(A) $XeO_2$(+4) and $XeO_3$(+6)

(B) $XeOF_4$(+6) and $XeO_3$(+6)

(C) $XeO_2F_2$(+6) and $XeO_2$(+4)

(D) $XeOF_4$(+6) and $XeO_2F_2$(+6)

Open solution

**Q228.** JEE Main 2017 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

$XeF_6$ on partial hydrolysis with water produces a compound 'X'. The same compound 'X' is formed when $XeF_6$ reacts with silica. The compound 'X' is :

(A) $XeF_2$

(B) $XeF_4$

(C) $XeOF_4$

(D) $XeO_3$

Open solution

**Q229.** JEE Main 2017 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

The number of P–OH bonds and the oxidation state of phosphorus atom in pyrophosphoric acid ($H_4P_2O_7$) respectively are :

(A) four and four

(B) five and four

(C) five and five

(D) four and five

Open solution

**Q230.** JEE Main 2017 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

The correct sequence of decreasing number of $\pi$-bonds in the structures of $H_2SO_3$, $H_2SO_4$ and $H_2S_2O_7$ is :

(A) $H_2SO_3 > H_2SO_4 > H_2S_2O_7$

(B) $H_2SO_4 > H_2S_2O_7 > H_2SO_3$

(C) $H_2S_2O_7 > H_2SO_4 > H_2SO_3$

(D) $H_2S_2O_7 > H_2SO_3 > H_2SO_4$

Open solution

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**Q231.** JEE Main 2017 (Online) 8th April Morning Slot · MCQ (Single Correct Answer)

The number of S = O and S − OH bonds present in peroxodisulphuric acid and pyrosulphuric acid respectively are :

(A) (2 and 2) and (2 and 2)
(B) (2 and 4) and (2 and 4)
(C) (4 and 2) and (2 and 4)
(D) (4 and 2) and (4 and 2)

Open solution

**Q232.** JEE Main 2017 (Offline) · MCQ (Single Correct Answer)

The products obtained when chlorine gas reacts with cold and dilute aqueous NaOH are :

(A) $Cl^-$ and $ClO_2^-$
(B) $ClO^-$ and $ClO_3^-$
(C) $Cl^-$ and $ClO^-$
(D) $ClO_2^-$ and $ClO_3^-$

Open solution

**Q233.** JEE Main 2016 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

**Assertion :** Among the carbon allotropes, diamond is an insulator, whereas, graphite is a good conductor of electricity.

**Reason :** Hybridization of carbon in diamond and graphite are $sp^3$ and $sp^2$, respectively.

(A) Both assertion and reason are correct, and the reason is the correct explanation for the assertion.
(B) Both assertion and reason are correct, but the reason is not the correct explanation for the assertion
(C) Assertion is incorrect statement, but the reason is correct.
(D) Both assertion and reason are incorrect.

Open solution

**Q234.** JEE Main 2016 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

Identify the **incorrect** statement :

(A) $S_2$ is paramagnetic like oxygen.
(B) Rhombic and monoclinic sulphur have $S_8$ molecules.
(C) $S_8$ ring has a crown shape.
(D) The S-S-S bond angles in the $S_8$ and $S_6$ rings are the same.

Open solution

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**Q235.** JEE Main 2016 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

Match the items in **Column I** with its main use listed in **Column II** :

Open solution

| Column I | Column II |
|---|---|
| (A) Silica gel | (i) Transistor |
| (B) Silicon | (ii) Ion-exchanger |
| (C) Silicone | (iii) Drying agent |
| (D) Silicate | (iv) Sealant |

**(A)** (A)-(iii), (B)-(i), (C)-(iv), (D)-(ii)
**(B)** (A)-(iv), (B)-(i), (C)-(ii), (D)-(iii)
**(C)** (A)-(ii), (B)-(iv), (C)-(i), (D)-(iii)
**(D)** (A)-(ii), (B)-(i), (C)-(iv), (D)-(iii)

**Q236.** JEE Main 2016 (Offline) · MCQ (Single Correct Answer)

The pair in which phosphorous atoms have a formal oxidation state of +3 is :

Open solution

**(A)** Orthophosphorous and pyrophosphorous acids
**(B)** Pyrophosphorous and hypophosphoric acids
**(C)** Orthophosphorus and hypophosphoric acids
**(D)** Pyrophosphorous and pyrophosphoric acids

**Q237.** JEE Main 2015 (Online) 11th April Morning Slot · MCQ (Single Correct Answer)

Which of the following compounds has a P – P bond ?

Open solution

**(A)** $H_4P_2O_5$ **(B)** $H_4P_2O_6$ **(C)** $H_4P_2O_7$ **(D)** $(HPO_3)_3$

**Q238.** JEE Main 2015 (Online) 11th April Morning Slot · MCQ (Single Correct Answer)

Chlorine water on standing loses its colour and forms :

Open solution

**(A)** HCl only **(B)** HOCl and $HOCl_2$ **(C)** HCl and HOCl **(D)** HCl and $HClO_2$

**Q239.** JEE Main 2015 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

The least number of oxyacids are formed by :

Open solution

**(A)** Chlorine **(B)** Fluorine **(C)** Sulphur **(D)** Nitrogen

**Q240.** JEE Main 2015 (Offline) · MCQ (Single Correct Answer)

Which among the following is the most reactive?

Open solution

**(A)** $Br_2$ **(B)** $I_2$ **(C)** ICl **(D)** $Cl_2$

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**Q241.** JEE Main 2014 (Online) 19th April Morning Slot · MCQ (Single Correct Answer)

Which of these statements is **not** true ?

(A) $NO^+$ is not isoelectronic with $O_2$

(B) B is always covalent in its compounds

(C) In aqueous solution, the $Tl^+$ ion is much more stable than Tl (III)

(D) $LiAlH_4$ is a versatile reducing agent in organic synthesis.

Open solution

**Q242.** JEE Main 2014 (Online) 12th April Morning Slot · MCQ (Single Correct Answer)

Which of the following xenon-OXO compounds may not be obtained by hydrolysis of xenon fluorides ?

(A) $Xe\,O_2F_2$ (B) $Xe\,O\,F_4$ (C) $Xe\,O_3$ (D) $Xe\,O_4$

Open solution

**Q243.** JEE Main 2014 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

In the following sets of reactants which two sets best exhibit the amphoteric character of $Al_2O_3 . xH_2O$ ?

Set 1 : $Al_2O_3 . xH_2O$ (s) and $OH^-$ (aq)

Set 2 : $Al_2O_3 . xH_2O$ (s) and $H_2O$ ($\ell$)

Set 3 : $Al_2O_3 . xH_2O$ (s) and $H^+$ (aq)

Set 4 : $Al_2O_3 . xH_2O$ (s) and $NH_3$ (aq)

(A) 1 and 2 (B) 1 and 3 (C) 2 and 4 (D) 3 and 4

Open solution

**Q244.** JEE Main 2014 (Offline) · MCQ (Single Correct Answer)

Among the following oxoacids, the correct decreasing order of acid strength is :

(A) $HClO_4 > HClO_3 > HClO_2 > HOCl$

(B) $HClO_2 > HClO_4 > HClO_3 > HOCl$

(C) $HOCl > HClO_2 > HClO_3 > HClO_4$

(D) $HClO_4 > HOCl > HClO_3 > HClO_2$

Open solution

**Q245.** JEE Main 2013 (Online) 22th April Morning Slot · MCQ (Single Correct Answer)

$XeO_4$ molecule is tetrahedral having:

(A) Two $p\pi$ - $d\pi$ bonds

(B) One $p\pi$ - $d\pi$ bonds

(C) Four $p\pi$ - $d\pi$ bonds

(D) Three $p\pi$ - $d\pi$ bonds

Open solution

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**Q246.** AIEEE 2012 (Online) 19th May · MCQ (Single Correct Answer)

The compound of xenon with zero dipole moment is

(A) $XeO_3$ (B) $XeF_4$ (C) $XeOF_4$ (D) $XeO_2$

Open solution

**Q247.** AIEEE 2012 (Online) 7th May · MCQ (Single Correct Answer)

In which of the following arrangements, the sequence is not strictly according to the property written against it?

(A) $CO_2 < SiO_2 < SnO_2 < PbO_2$ : increasing oxidising power

(B) $NH_3 < PH_3 < AsH_3 < SbH_3$ : increasing basic strength

(C) $HF < HCl < HBr < HI$ : increasing acid strength

(D) $B < C < O < N$ : increasing first ionisation enthalpy

Open solution

**Q248.** AIEEE 2011 · MCQ (Single Correct Answer)

Boron cannot form which one of the following anions?

(A) $BH_4^-$ (B) $BF_6^{3-}$ (C) $B(OH)_4^-$ (D) $BO_2^-$

Open solution

**Q249.** AIEEE 2009 · MCQ (Single Correct Answer)

Which one of the following reactions of Xenon compounds is not feasible?

(A) $XeO_3 + 6HF \rightarrow XeF_6 + 3H_2O$

(B) $XeF_6 + RbF \rightarrow Rb(XeF_7]$

(C) $2XeF_2 + 2H_2O \rightarrow 2Xe + 4HF + O_2$

(D) $3XeF_4 + 6H_2O \rightarrow 2Xe + XeO_3 + 12HF + 1.5O_2$

Open solution

**Q250.** AIEEE 2008 · MCQ (Single Correct Answer)

Among the following substituted silanes the one which will give rise to cross linked silicone polymer on hydrolysis is :

(A) $R_4Si$ (B) $RSiCl_3$ (C) $R_2SiCl_2$ (D) $R_3SiCl$

Open solution

**Q251.** AIEEE 2007 · MCQ (Single Correct Answer)

Identify the incorrect statement among the following :

(A) Ozone reacts with $SO_2$ to give $SO_3$

(B) Silicon reacts with $NaOH_{(aq)}$ in the presence of air to give $Na_2SiO_3$ and $H_2O$

(C) $Cl_2$ reacts with excess of $NH_3$ to give $N_2$ and HCl

(D) $Br_2$ reacts with hot and strong NaOH solution to give NaBr, $NaBrO_4$ and $H_2O$

Open solution

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**Q252.** AIEEE 2006 · MCQ (Single Correct Answer)

What products are expected from the disproportionation reaction of hypochlorous acid?

(A) $HClO_3$ and $Cl_2O$ (B) $HClO_2$ and $HClO_4$ (C) HCl and $Cl_2O$ (D) HCl and $HClO_3$

Open solution

**Q253.** AIEEE 2005 · MCQ (Single Correct Answer)

The number of hydrogen atom(s) attached to phosphorus atom in hypophosphorous acid is :

(A) zero (B) two (C) one (D) three

Open solution

**Q254.** AIEEE 2005 · MCQ (Single Correct Answer)

The correct order of the thermal stability of hydrogen halides (H – X) is :

(A) HI > HBr > HCl > HF (B) HF > HCl > HBr > HI

(C) HCl < HF > HBr < HI (D) HI > HCl < HF < HBr

Open solution

**Q255.** AIEEE 2005 · MCQ (Single Correct Answer)

Heating an aqueous solution of aluminium chloride to dryness will give :

(A) $AlCl_3$ (B) $Al_2Cl_6$ (C) $Al_2O_3$ (D) $Al(OH)Cl_2$

Open solution

**Q256.** AIEEE 2005 · MCQ (Single Correct Answer)

In silicon dioxide :

(A) Each silicon atom is surrounded by four oxygen atoms and each oxygen atom is bonded to two silicon atoms

(B) Each silicon atom is surrounded by two oxygen atoms and each oxygen atom is bonded to two silicon atoms

(C) Silicon atoms is bonded to two oxygen atoms

(D) there are double bonds between silicon and oxygen atoms

Open solution

**Q257.** AIEEE 2005 · MCQ (Single Correct Answer)

The structure of diborane ($B_2H_6$) contains :

(A) four 2c-2e bonds and two 3c-2e bonds (B) two 2c-2e bonds and four 3c-2e bonds

(C) two 2c-2e bonds and two 3c-3e bonds (D) four 2c-2e bonds and four 3c-2e bonds

Open solution

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**Q258.** AIEEE 2004 · MCQ (Single Correct Answer)

Which among the following factors is the most important in making fluorine the strongest oxidizing halogen?

(A) Electron affinity

(B) Bond dissociation energy

(C) Hydration enthalpy

(D) Ionization enthalpy

Open solution

**Q259.** AIEEE 2004 · MCQ (Single Correct Answer)

Aluminium chloride exists as dimer, $Al_2Cl_6$ in solid state as well as in solution of non-polar solvents such as benzene. When dissolved in water, it gives :

(A) $Al3^+ + 3Cl^-$

(B) $Al_2O_3 + 6HCl$

(C) $[Al(OH)_6]^{3-}$

(D) $[Al(H_2O)_6]^{3+} + 3Cl^-$

Open solution

**Q260.** AIEEE 2004 · MCQ (Single Correct Answer)

The soldiers of Napolean army while at Alps during freezing winter suffered a serious problem as regards to the tin buttons of their uniforms. White metallic tin buttons got converted to grey power. The transformation is related to :

(A) a change in the partial pressure of oxygen in the air

(B) a change in the crystalline structure of tin

(C) an interaction with nitrogen of the air at very low temperature

(D) an interaction with water vapor contained in the humid air

Open solution

**Q261.** AIEEE 2003 · MCQ (Single Correct Answer)

What may be expected to happen when phosphine gas is mixed with chlorine gas?

(A) $PCl_3$ and HCl are formed and the mixture warms up

(B) $PCl_5$ and HCl are formed and the mixture cools down

(C) $PH_3.Cl_2$ is formed with warming up

(D) The mixture only cools down

Open solution

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**Q262.** AIEEE 2003 · MCQ (Single Correct Answer)

Concentrated hydrochloric acid when kept in open air sometimes produces a cloud of white fumes. The explanation for it is that :

(A) oxygen in air reacts with the emitted HCl gas to form a cloud of chlorine gas

(B) strong affinity of HCl gas for moisture in air results in forming of droplets of liquid solution which appears like a cloudy smoke

(C) due to strong affinity for water, concentrated hydrochloric acid pulls moisture of air towards it self. This moisture forms droplets of water and hence the cloud.

(D) concentrated hydrochloric acid emits strongly smelling HCl gas all the time.

Open solution

**Q263.** AIEEE 2003 · MCQ (Single Correct Answer)

Graphite is a soft solid lubricant extremely difficult to melt. The reason for this anomalous behaviour is that graphite :

(A) is an allotropic form of diamond

(B) has molecules of variable molecular masses like polymers

(C) has carbon atoms arranged in large plates of rings of strongly bound carbon atoms with weak interplate bonds

(D) is a non-crystalline substance

Open solution

**Q264.** AIEEE 2002 · MCQ (Single Correct Answer)

In $XeF_2$, $XeF_4$, $XeF_6$ the number of lone pairs of Xe are respectively :

(A) 2, 3, 1 (B) 1, 2, 3 (C) 4, 1, 2 (D) 3, 2, 1

Open solution

**Q265.** AIEEE 2002 · MCQ (Single Correct Answer)

In case of nitrogen, $NCl_3$ is possible but not NCl5 while in case of phosphorous, $PCl_3$ as well as $PCl_5$ are possible. It is due to :

(A) availability of vacant d orbitals in P but not in N

(B) lower electronegativity of P than N

(C) lower tendency of H - bond formation in P than N

(D) occurrence of P in solid while N in gaseous state at room temperature

Open solution

**Q266.** AIEEE 2002 · MCQ (Single Correct Answer)

Number of sigma bonds in $P_4O_{10}$ is :

(A) 6 (B) 7 (C) 17 (D) 16

Open solution

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**Q267.** AIEEE 2002 · MCQ (Single Correct Answer)

When $H_2S$ is passed through $Hg_2S$ we get :

(A) HgS (B) HgS + $Hg_2S$ (C) $Hg_2S$ (D) $Hg_2S_2$

Open solution

**Q268.** JEE Main 2023 (Online) 13th April Evening Shift · Numerical

If the formula of Borax is $Na_2B_4O_x(OH)_y \cdot zH_2O$, then $x + y + z =$ _____

Open solution

**Q269.** JEE Main 2023 (Online) 10th April Evening Shift · Numerical

The difference in the oxidation state of $Xe$ between the oxidised product of $Xe$ formed on complete hydrolysis of $XeF_4$ and $XeF_4$ is _____

Open solution

**Q270.** JEE Main 2023 (Online) 8th April Evening Shift · Numerical

The ratio of sigma and $\pi$ bonds present in pyrophosphoric acid is _____.

Open solution

**Q271.** JEE Main 2023 (Online) 8th April Morning Shift · Numerical

$XeF_4$ reacts with $SbF_5$ to form

$[XeF_m]^{n+}[SbF_y]^{z-}$.

$m + n + y + z =$ _____

Open solution

**Q272.** JEE Main 2022 (Online) 29th July Evening Shift · Numerical

Consider the following sulphur based oxoacids.

$H_2SO_3, H_2SO_4, H_2S_2O_8$ and $H_2S_2O_7$.

Amongst these oxoacids, the number of those with peroxo $(O-O)$ bond is _____.

Open solution

**Q273.** JEE Main 2022 (Online) 26th July Evening Shift · Numerical

The number of non-ionisable protons present in the product B obtained from the following reactions is _____.

$C_2H_5OH + PCl_3 \rightarrow C_2H_5Cl + A$

$A + PCl_3 \rightarrow B$

Open solution

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**Q274.** JEE Main 2022 (Online) 25th June Evening Shift · **Numerical**

Consider the following reactions:

$PCl_3 + H_2O \rightarrow A + HCl$

$A + H_2O \rightarrow B + HCl$

The number of ionisable protons present in the product B is _____.

Open solution

**Q275.** JEE Main 2021 (Online) 31st August Morning Shift · **Numerical**

Consider the sulphides HgS, PbS, CuS, $Sb_2S_3$, $As_2S_3$ and CdS. Number of these sulphides soluble in 50% $HNO_3$ is _____.

Open solution

**Q276.** JEE Main 2021 (Online) 31st August Morning Shift · **Numerical**

The number of halogen/(s) forming halic (V) acid is _____.

Open solution

**Q277.** JEE Main 2021 (Online) 18th March Evening Shift · **Numerical**

A xenon compound 'A' upon partial hydrolysis gives $XeO_2F_2$. The number of lone pair of electrons present in compound A is _____. (Round off to the Nearest Integer)

Open solution

**Q278.** JEE Main 2021 (Online) 17th March Morning Shift · **Numerical**

The reaction of white phosphorus on boiling with alkali in inert atmosphere resulted in the formation of product 'A'. The reaction of 1 mol of 'A' with excess of $AgNO_3$ in aqueous medium gives _____ mol(s) of Ag. (Round off to the Nearest Integer).

Open solution

**Q279.** JEE Main 2021 (Online) 24th February Evening Shift · **Numerical**

Among the following allotropic forms of sulphur, the number of allotropic forms, which will show paramagnetism is _____.

(A) $\alpha$-sulphur

(B) $\beta$-sulphur

(C) $S_2$-form

Open solution

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**Q280.** JEE Main 2013 (Offline) · **MCQ (More than One Correct Answer)**

Which of the following is the wrong statement?

(A) $O_3$ molecule is bent

(B) Ozone is violet-black in solid state

(C) Ozone is diamagnetic gas

(D) ONCl and $ONO^-$ are isoelectronic

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# Answer Key

## MCQ (Single Correct Answer)

| | | | | |
|---|---|---|---|---|
| 1. **C** | 2. **A** | 3. **A** | 4. **D** | 5. **D** |
| 6. **C** | 7. **A** | 8. **A** | 9. **B** | 10. **D** |
| 11. **A** | 12. **C** | 13. **B** | 14. **B** | 15. **B** |
| 16. **D** | 17. **A** | 18. **C** | 19. **C** | 20. **A** |
| 21. **A** | 22. **A** | 23. **C** | 24. **C** | 25. **C** |
| 26. **A** | 27. **B** | 28. **B** | 29. **C** | 30. **D** |
| 31. **D** | 32. **D** | 33. **B** | 34. **D** | 35. **D** |
| 36. **B** | 37. **C** | 38. **B** | 39. **B** | 40. **B** |
| 41. **A** | 42. **D** | 43. **C** | 44. **B** | 45. **A** |
| 46. **A** | 47. **C** | 48. **A** | 49. **C** | 50. **C** |
| 51. **D** | 52. **B** | 53. **D** | 54. **B** | 55. **A** |
| 56. **B** | 57. **D** | 58. **A** | 59. **D** | 60. **A** |
| 61. **C** | 62. **D** | 63. **B** | 64. **C** | 65. **D** |
| 66. **D** | 67. **B** | 68. **A** | 69. **D** | 70. **B** |
| 71. **A** | 72. **B** | 73. **A** | 74. **D** | 75. **A** |
| 76. **C** | 77. **B** | 78. **A** | 79. **A** | 80. **C** |
| 81. **C** | 82. **D** | 83. **B** | 84. **D** | 85. **C** |
| 86. **D** | 87. **A** | 88. **B** | 89. **D** | 90. **C** |
| 91. **A** | 92. **A** | 93. **C** | 94. **A** | 95. **B** |
| 96. **D** | 97. **A** | 98. **C** | 99. **D** | 100. **D** |
| 101. **A** | 102. **C** | 103. **D** | 104. **C** | 105. **C** |
| 106. **D** | 107. **A** | 108. **B** | 109. **C** | 110. **C** |
| 111. **D** | 112. **C** | 113. **C** | 114. **D** | 115. **D** |
| 116. **D** | 117. **A** | 118. **A** | 119. **A** | 120. **B** |
| 121. **D** | | | | |

## Numerical

| | | | | |
|---|---|---|---|---|
| 122. **15** | 123. **7** | 124. **3** | 125. **8** | 126. **3** |
| 127. **5** | 128. **8** | 129. **12** | 130. **3** | |

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| | | | | |
|---|---|---|---|---|
| 131. **C** | 132. **D** | 133. **D** | 134. **B** | 135. **B** |
| 136. **C** | 137. **D** | 138. **A** | 139. **C** | 140. **B** |
| 141. **D** | 142. **C** | 143. **D** | 144. **D** | 145. **A** |
| 146. **B** | 147. **C** | 148. **B** | 149. **C** | 150. **B** |
| 151. **D** | 152. **D** | 153. **B** | 154. **A** | 155. **B** |
| 156. **B** | 157. **C** | 158. **B** | 159. **B** | 160. **D** |
| 161. **A** | 162. **B** | 163. **C** | 164. **A** | 165. **D** |
| 166. **B** | 167. **D** | 168. **D** | 169. **A** | 170. **B** |
| 171. **A** | 172. **C** | 173. **B** | 174. **D** | 175. **D** |
| 176. **D** | 177. **D** | 178. **B** | 179. **A** | 180. **B** |
| 181. **A** | 182. **D** | 183. **B** | 184. **B** | 185. **A** |
| 186. **C** | 187. **A** | 188. **B** | 189. **B** | 190. **C** |
| 191. **D** | 192. **A** | 193. **D** | 194. **A** | 195. **B** |
| 196. **C** | 197. **A** | 198. **D** | 199. **B** | 200. **A** |
| 201. **B** | 202. **C** | 203. **D** | 204. **D** | 205. **B** |
| 206. **B** | 207. **D** | 208. **C** | 209. **A** | 210. **B** |
| 211. **B** | 212. **D** | 213. **D** | 214. **D** | 215. **C** |
| 216. **D** | 217. **D** | 218. **D** | 219. **C** | 220. **D** |
| 221. **D** | 222. **A** | 223. **D** | 224. **C** | 225. **B** |
| 226. **C** | 227. **D** | 228. **C** | 229. **D** | 230. **C** |
| 231. **D** | 232. **C** | 233. **B** | 234. **D** | 235. **A** |
| 236. **A** | 237. **B** | 238. **C** | 239. **B** | 240. **C** |
| 241. *Scan QR* | 242. **D** | 243. **B** | 244. **A** | 245. **C** |
| 246. **B** | 247. **B** | 248. **B** | 249. **A** | 250. **B** |
| 251. **D** | 252. **D** | 253. **B** | 254. **B** | 255. **C** |
| 256. **A** | 257. **A** | 258. **B** | 259. **D** | 260. **B** |
| 261. **A** | 262. **A** | 263. **C** | 264. **D** | 265. **A** |
| 266. **D** | 267. **B** | 268. **17** | 269. **2** | 270. **6** |
| 271. **11** | 272. **1** | 273. **2** | 274. **2** | 275. **4** |
| 276. **3** | 277. **19** | 278. **4** | 279. **1** | 280. **D** |

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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Practise with the questions your exam has already asked, then test yourself under real exam conditions.

PAST YEAR QUESTIONS

## Every past paper, chapter-wise.

Previous year questions sorted the way you study, each with a full solution.

- Chapter-wise and year-wise
- Weightage and topic tags on every chapter
- Your progress on each one, at a glance

MOCK TESTS

## Mock tests with a real exam feel.

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- Mark questions for review and come back

ANALYSIS

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A full report after every test, so you know exactly where you stand.

- Your percentile among everyone who took it
- Predicted real-exam percentile and All India Rank
- Subject breakdown and time analysis

PRACTICE

## Solve. Check. Understand.

Learn from every question, not just the ones you get right.

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- Step-by-step explanations
- See how everyone else answered

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With full solutions

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Subject and chapter-wise

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